MIL-T-81695A(AS)
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MIL-T-81695(AS)
1 September 1972

MILITARY SPECIFICATION

TRANSMITTING SET, TELEMETRIC DATA, AN/AKT-22(V) (U)

This specification has been approved

by the

Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 (U) Scope - The equipment covered by this specification is an airborne ASW-signal transmitting set which provides a complete data link between acoustic signal acquisition sonobuoys and a shipboard data processing system for both sonic and sonar data. The AN/AKT-22(V) is a variable configuration device for multiple airframe use. By addition or deletion of modules the output power, multiplex format and/or channel capacity may be changed to meet the requirements of the specific airframe.

*1.2 (U) Classification - The equipment covered by this specification shall consist of the following items.

Type Designation	Applicable Paragraph
Transmitter-Multiplexer T-1220/AKT-22(V)	3.5.1 AN/AKT-22(V)1
Transmitter-Multiplexer T-1220A/AKT-22(V)	3.5.2 AN/AKT-22(V)2
Transmitter-Multiplexer T-1295/AKT-22(V)	3.5.3 AN/AKT-22(V)3
Transmitter-Multiplexer T-1220B/AKT-22(V)	3.5.4 AN/AKT-22(V)4
Antenna Coupler CU-1497/ARR	3.5.5 AN/AKT-22(V)1
Antenna Coupler CU-2069/ARR	3.5.6 AN/AKT-22(V)1, 2, 3, or 4
Antenna AS-2744/AKT-22(V)	3.5.7 AN/AKT-22(V)1 or 2
Antenna AS-3033/AKT-22(V)	3.5.8 AN/AKT-22(V)3, or 4
Actuator, Linear Electro-Mechanical TG-229/AKT-22(V)	3.5.9 AN/AKT-22(V)3, or 4
Control-Indicator C-8988/AKT-22(V)	3.5.10 AN/AKT-22(V)1 or 3
Control-Indicator C-8988A/AKT-22(V)	3.5.11 AN/AKT-22(V)2 or 4
Amplifier AM-6695/AKT-22(V)	3.5.12 AN/AKT-22(V)2 or 4

COMMENTS OR RECOMMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY.

FSC 5821

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1.3 (U) Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 (U) General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirements for

STANDARDS

Military

MIL-STD-188	Military Communications System Technical Standards
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Nonexpendable Electronic Equipment
MIL-STD-785	Reliability Program for System and Equipment Development and Production

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STANDARDS

Military

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

MS 91403 Cases, Large Size (For use with electronic equipment in aircraft)

2.1.1 (U) Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 (U) Preproduction - This specification makes provision for preproduction testing.

3.2 (U) Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 (U) Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 (U) Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

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3.2.3 (U) Modules - The electronic portions of the equipment shall be functionally modularized in accordance with AR-10.

3.3 (U) Design and Construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 (U) Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and the configuration utilized. It shall not exceed 46 pounds.

3.3.2 (U) Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 (U) Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 250 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

*3.3.2.2 (U) Operating Life - The equipment shall have a total operating life of 15,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

*3.3.2.3 (U) Reliability in Mean Time Between Failure (MTBF) - The equipment, shall have 250 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 (U) Time Totalizing Meter - A time totalizing meter conforming to MIL-M-7793 shall be included in the transmitter-multiplexer.

3.3.3 (U) Cabling and Connections -

3.3.3.1 (U) Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 (U) Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 (U) Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

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3.3.5 (U) Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400.

*3.3.5.1 (U) Unit Interchangeability -

a) All common plug-in units used in T-1220, T-1220A, T-1295, and T-1220B shall be interchangeable.

b) The following interchangeability among units shall be required

1. CU-2069/ARR may always replace CU-1497/ARR.
2. The linear actuator of AS-2744 shall be identical to TG-229/AKT-22.
3. Except for the specified gain, beamwidth, and envelope antennas AS-2744 and AS-3033 shall be interchangeable.
4. Except for plug-in complement the T-1295 and T-1220B shall be identical and interchangeable.
5. Except for specified power output and plug-in complement T-1220 and T-1220A shall be identical and interchangeable.
6. The T-1295 may directly replace the T-1220 and the T-1220B may directly replace the T-1220A.

*3.3.6 (U) Interference Control. - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461A, Notice 1. Existing designs are permitted to 20 db degradation from the levels called out in MIL-STD-461A, Notice 1.

3.3.7 (U) Provisions for Maintainability - The Maintainability Program, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in AR-10.

3.3.8 (U) Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

*3.3.9 (U) Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidty
Input power voltage	115/200 volts 400 ±20 Hz 3-phase Y and 28.0 ±0.5 volts dc

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*3.3.10 (U) Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1A equipment except as modified herein.

*3.3.10.1 (U) Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of the curve shown in Figure 32, without isolators.

*3.3.10.2 (U) Altitude - The equipment shall operate satisfactorily at any altitude up to 10,000 feet.

3.3.11 (U) Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.3.12 (U) Input Electrical Power -

*3.3.12.1 (U) Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC Power (Three Phase Wye), 115/200 volts, 400 Hz, Category A, 300VA for AN/AKT-22(V)1 and AN/AKT-22(V)2 and 500 VA for AN/AKT-22(V)3 and AN/AKT-22(V)4.

(2) DC Power, 28 V, Category A, 50 watts, maximum

3.3.12.2 (U) Lighting Power - Input power for panel lighting shall require not more than 0.15 amp at 28 volts.

*3.3.12.3 (U) Degraded Performance - Degraded performance will be permitted for voltage transients on the AC buss per MIL-STD-704A. Operation shall return to normal with no resulting damage to the equipment.

*3.3.12.4 (U) Standby Power - The equipment shall provide for a standby position whereby rated voltage shall be applied to all circuits except the transmitter. The standby power shall not exceed 300 VA ac power, 50 watts dc power, and 0.15 amp lighting power.

3.4 (U) Performance - Unless otherwise specified, values set forth to establish the requirements of satsifactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions are specified herein.

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*3.4.1 (U) Operation - When properly interconnected to a suitable airborne sonobuoy receiver such as the AN/ARR-52A or the AN/ARR-75, Telemetric Data Transmitting Sets AN/AKT-22(V) shall transpond signals from selected sonobuoys to a shipboard receiving set and acoustic processor. Block diagrams of the AN/AKT-22(V)1, AN/AKT-22(V)2, AN/AKT-22(V)3, and AN/AKT-22(V)4 are presented in figures 1, 2, 3, and 4, respectively. The transmitting sets shall operate with sonobuoys and telemetric data receiving sets as follows: x indicates operation with specified equipment
L indicates operation with LOFAR component only
f indicates operation with Limited RF frequencies
c indicates operation with Reduced Sonobuoy Channels

		Telemetric Data Transmitting Set			
		AN/AKT-22(V)1	AN/AKT-22(V)2	AN/AKT-22(V)3	AN/AKT-22(V)4
Sonobuoy	AN/SSQ-41	x	x	x	x
	AN/SSQ-47	x	x	x	x
	AN/SSQ-36	x	x	x	x
	AN/SSQ-53	L	x	L	x
Telemetric Data Receiving Set	AN/SKR-3A	x	x	f	f
	AN/SKR-3B	x	x	x	x
	AN/SKR-4	x	c	f	fc
	AN/SKR-4A	x	c	x	c
	AN/SKR-5	x	x	x	x

*3.4.2 (U) Operating Modes - AN/AKT-22(V)1, 2, 3, and 4 shall have off, standby, and operate modes and shall provide trigger control when AN/SSQ-47 Sonobuoys are used. AN/AKT-22(V)2 and 4 shall have a DIFAR mode to accommodate up to two AN/SSQ-53 Sonobuoys; the AN/AKT-22(V)1 and 3 shall be wired for the DIFAR mode. (See paragraphs 3.5.1.7.4 and 3.5.2.7.1.)

3.4.3 (U) Operating Frequency Range - AN/AKT-22(V)1 shall transmit over a frequency range from 2.200 to 2.245 GHz. AN/AKT-22(V)2, AN/AKT-22(V)3, and AN/AKT-22(V)4 shall transmit over a frequency range from 2.200 to 2.290 GHz.

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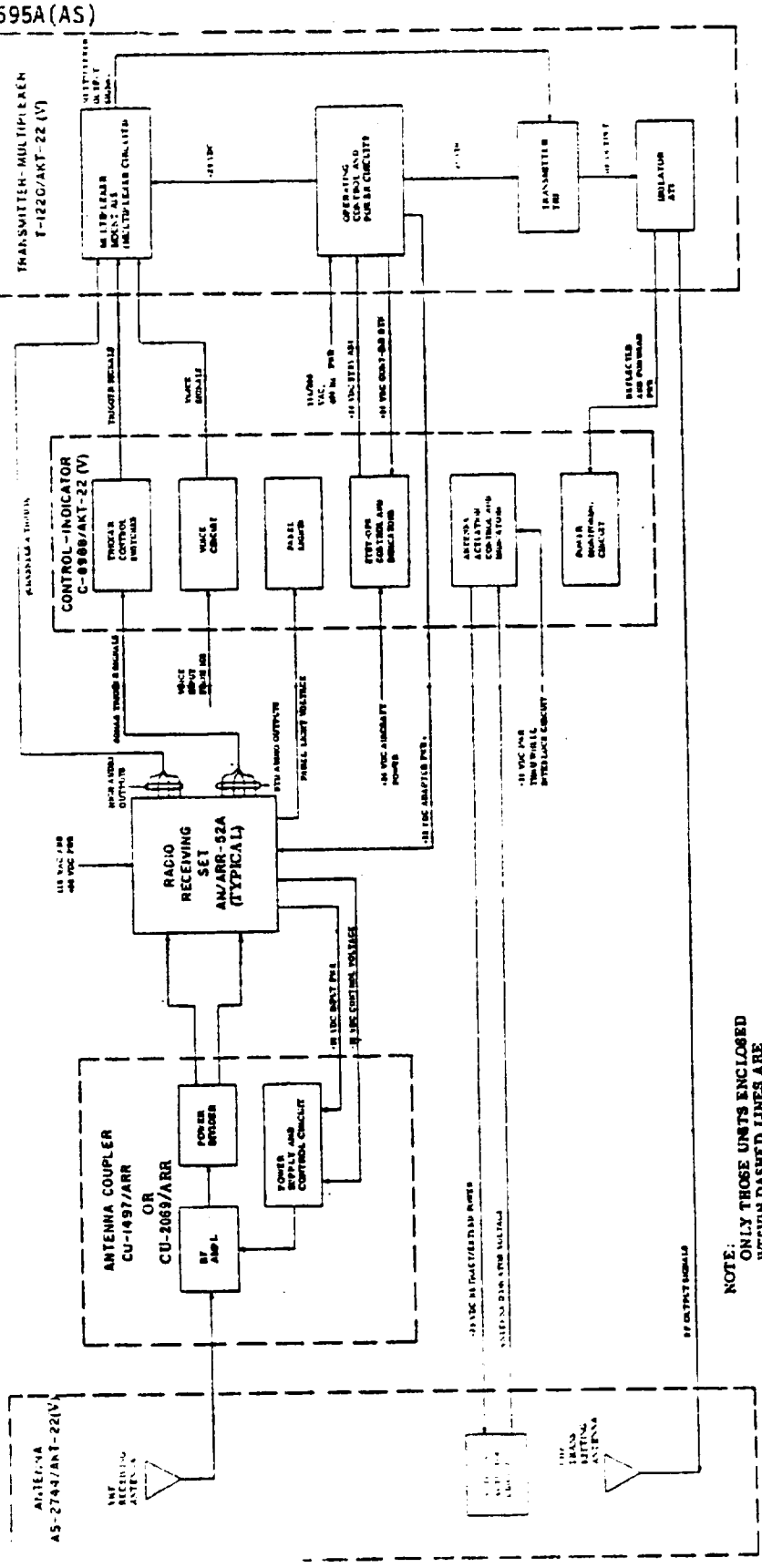


Figure 1. Telemetric Data Transmit Set AN/AKT-22(V)1, Block Diagram

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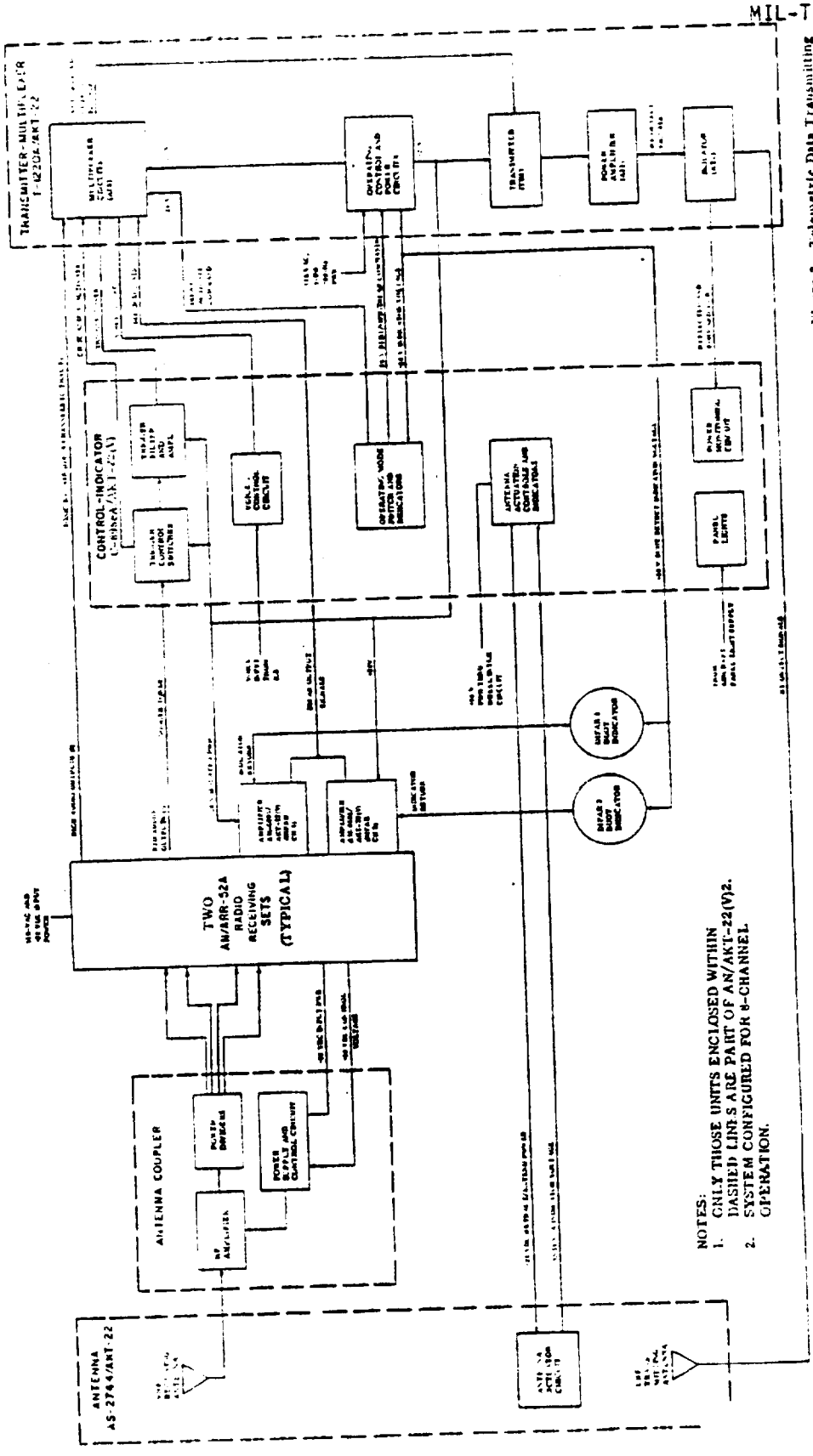


Figure 2. Telemetric Data Transmitting Set AN/AKT-22(V)2, Block Diagram

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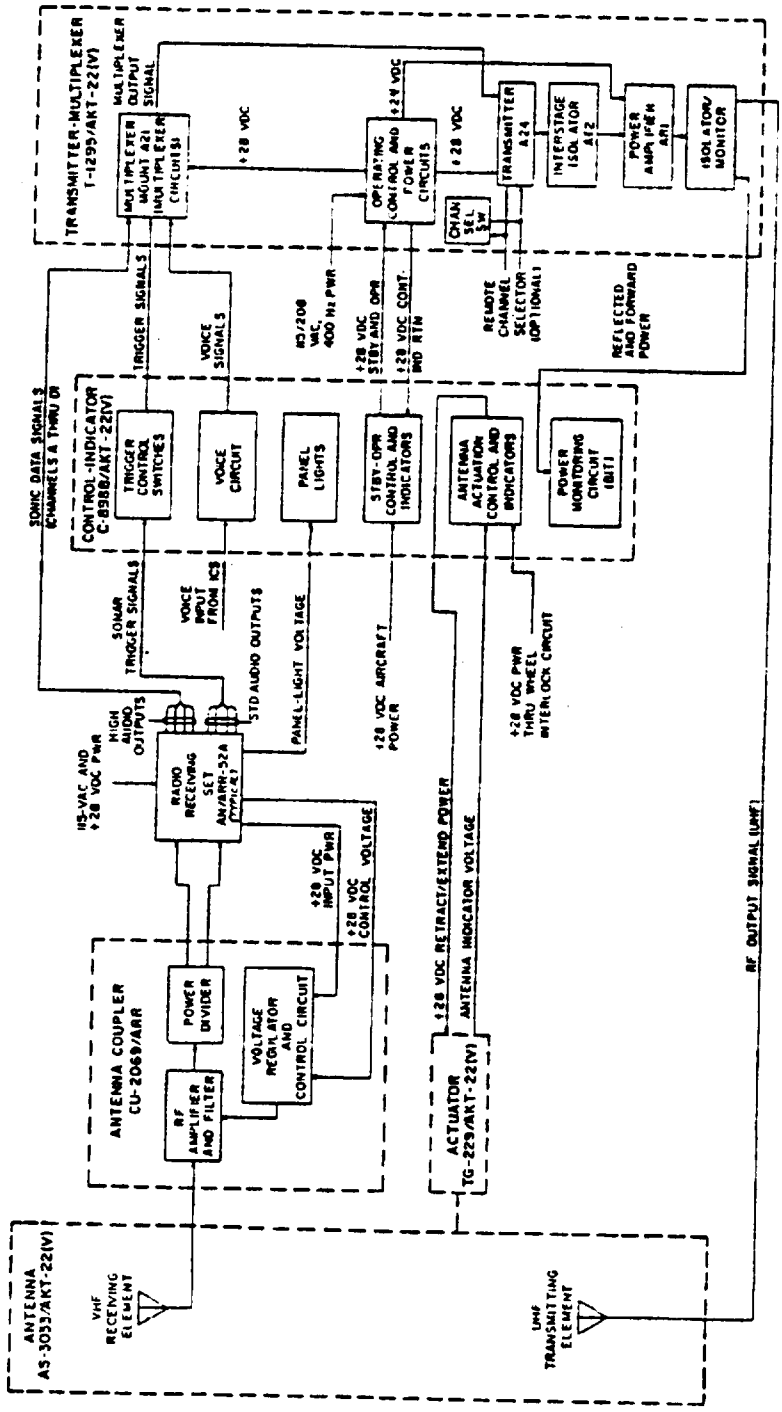


Figure 2. Telemetric Data Transmitting Set AN/AKT-22(V)3, Block Diagram

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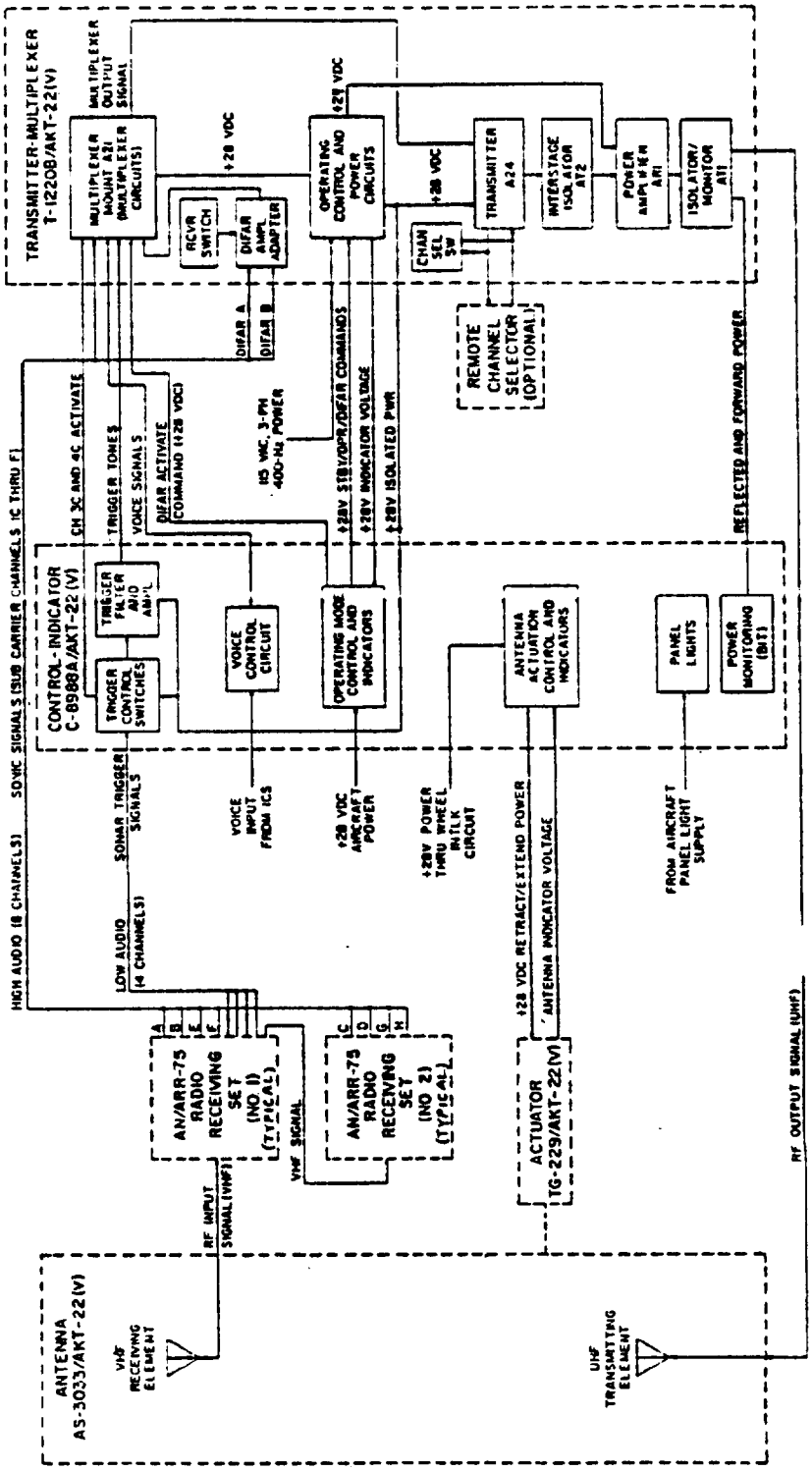


Figure 4. Telemetric Data Transmitting Set AN/AKT-22(V)4. Block Diagram

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*3.4.4 (U) Sonic Data Channel Output Performance - The sonic data channels shall meet the following performance requirements using the appropriate telemetric data receiving set (see paragraph 3.4.1).

*3.4.4.1 (U) Signal-to-Noise Ratio - The signal-to-noise ratio of each sonic data channel shall be 54 db minimum when measured in the range from 10 Hz to 100 kHz and 74 db minimum when measured for any 6 Hz bandwidth in the range from 10 Hz to 500 Hz.

*3.4.4.2 (U) Phase Correlation - Each sonic data channel shall have equal delay within ±5 degrees tracked over the band from 10 to 500 Hz in the 2-kHz bandwidth channels.

*3.4.4.3 (U) Frequency Response - Each sonic data channel shall have a frequency response within ±1.0 db from 10 Hz to 500 Hz; the response shall be -3 ±2 db at 2 kHz.

3.5 (U) Detail Requirements -

*3.5.1 (U) Transmitter-Multiplexer T-1220/AKT-22(V) - The transmitter-multiplexer shall meet the following requirements:

*3.5.1.1 (U) Function - The T-1220/AKT-22(V) transmitter-multiplexer shall condition and combine four sonic data channels, one voice channel, and up to four trigger signals, into a frequency-division multiplex signal and fm modulate a uhf carrier with the combined multiplex signal and amplify the carrier. The T-1220/AKT-22(V) shall be convertable to completely or partially perform the functions of the T-1220A/AKT-22(V) by the addition, replacement, or transfer of modules. Provision for modulation by an external video or PCM signals shall be provided. The transmitter-multiplexer shall be IMA repairable.

*3.5.1.2 (U) Form Factor - The transmitter-multiplexer shall be enclosed in a case with dimensions shown in figure 5.

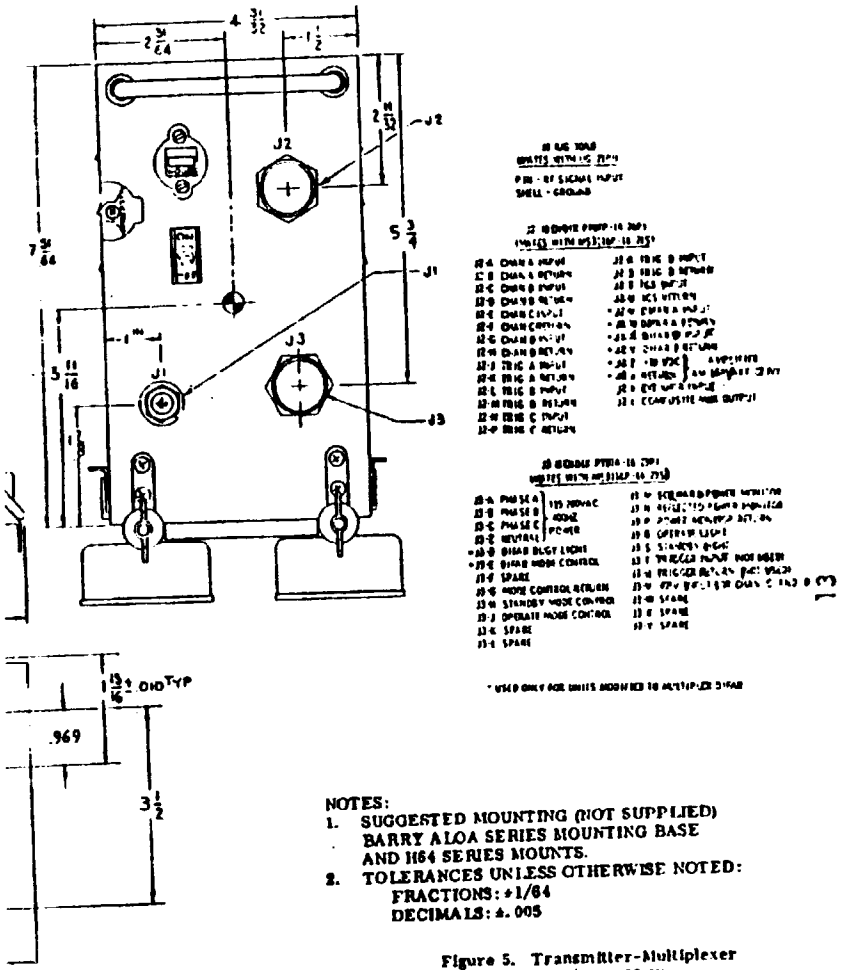
*3.5.1.3 (U) Weight - The weight of the transmitter-multiplexer shall not exceed 16.5 pounds.

*3.5.1.4 (U) Contents - The transmitter-multiplexer shall contain the following major assemblies:

- Four LOFAR voltage controlled oscillators per paragraph 3.5.1.8.
- Four LOFAR low-pass filters per paragraph 3.5.1.9
- One voice low-pass filter per paragraph 3.5.1.10
- One mixer-amplifier per paragraph 3.5.1.11
- One trigger filter per paragraph 3.5.1.12
- One fm transmitter per paragraph 3.5.1.13
- One isolator/power monitor per paragraph 3.5.1.14
- One telemetry mount per paragraph 3.5.1.15
- One internal power supply and necessary control circuitry.

*3.5.1.5 (U) Controls - The only operating control on the front panel of the transmitter-multiplexer shall be a circuit breaker for equipment protection.

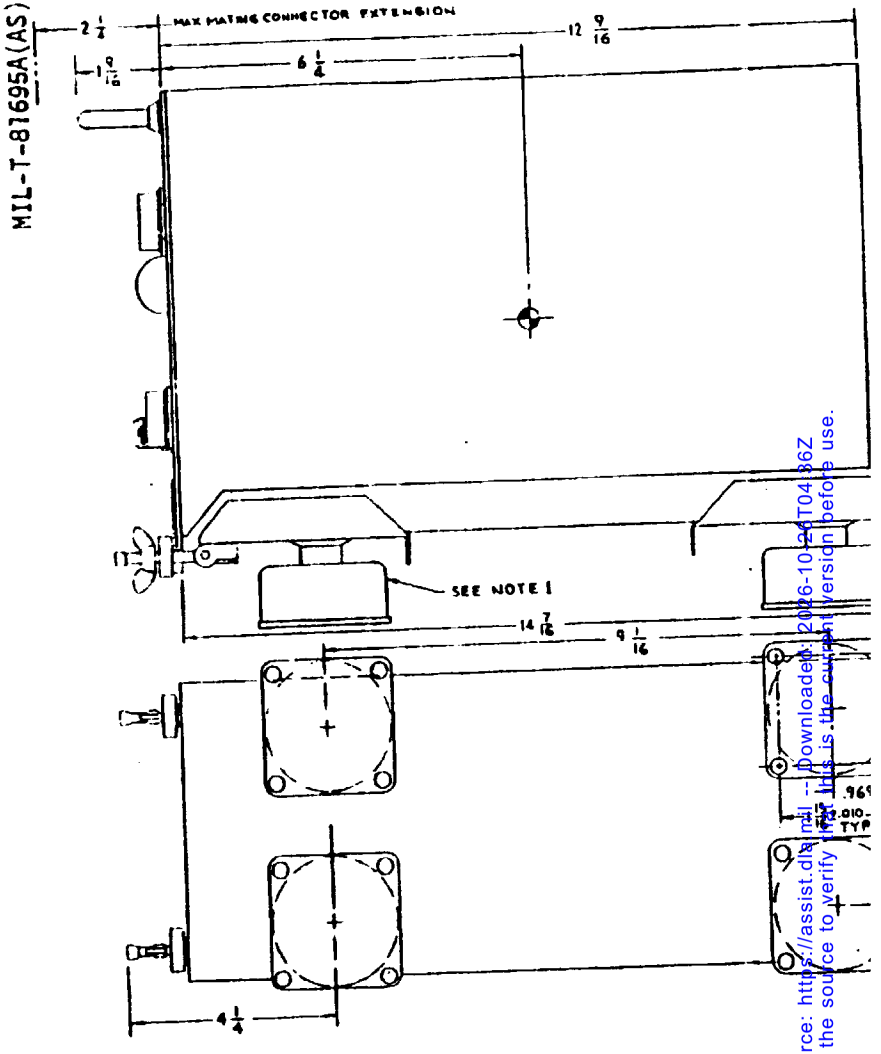
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NOTES:
1. SUGGESTED MOUNTING (NOT SUPPLIED) BARRY ALOA SERIES MOUNTING BASE AND H64 SERIES MOUNTS.
2. TOLERANCES UNLESS OTHERWISE NOTED:
   FRACTIONS: ±1/64
   DECIMALS: ±.005

Figure 5. Transmitter-Multiplexer T-1220/AKT-22(V) Form Factor and Connector Requirements

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*3.5.1.6 (U) Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	UG 30()/U (Type N)	RF Output
J2	PT07P-16-26P	Signal Inputs
J3	PT07A-16-23P	Power and Control

3.5.1.7 (U) Overall Unit - The overall transmitter-multiplexer shall meet the following requirements.

*3.5.1.7.1 (U) Primary Power Input - 111.5 to 116.5 volts average of 3 phases. 400 ±20 Hz, 3-phase.

*3.5.1.7.2 (U) Control Voltage Input - 25.0 to 28.5 volts dc.

*3.5.1.7.3 (U) Internal Regulated Power - 28 ±1.0 vdc.

*3.5.1.7.4 (U) Operating Modes -

a. STANDBY mode, shall turn on and apply power to multiplexer circuits.

b. OPERATE mode, shall apply operational power to transmitter. One voice channel and any four channels of data from AN/SSQ-41, AN/SSQ-47, or AN/SSQ-36 Sonobuoys or the LOFAR portion of data from an AN/SSQ-53 Sonobuoy shall be multiplexed and transmitted in this mode.

c. OFF mode, no power required.

d. A fourth mode shall be provided for DIFAR control (wired only).

*3.5.1.7.5 (C) Operating Frequency - One of the following as indicated by letter or color on transmitter and transmitter-multiplexer:

A - Red 2212.5 MHz

B - Green 2220.5 MHz

C - Yellow 2240.5 MHz

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*3.5.1.7.6 (C) Deviations - The deviations to the main carrier for the various subcarrier and baseband signals are:

Subcarrier (kHz)	Deviation (kHz)
12.8	±13.8
20.8	±22.5
45.8	±48.8
54.2	±58.8
Voice (baseband)	± 9.4
Trigger (each)	± 6.2

*3.5.1.7.7 (U) Frequency Accuracy - Within ±0.003 percent of specified unmodulated carrier frequency in temperature range from 0°C to +45°C; within 0.005 percent of specified unmodulated frequency in temperature ranges from -54°C to 0°C and +45°C to +55°C.

*3.5.1.7.8 (C) RF Power Output - 6.3 watts +2, -1 db into 50-ohm resistive load over frequency range of 2200 to 2245 MHz from -54°C to +55°C.

*3.5.1.7.9 (U) Signal Inputs -

a. Four sonic data channels with 10 to 2000 Hz bandwidth, 16.0 volts rms for full VCO deviation.

b. Four trigger channels with 26 to 38 kHz bandwidth, 1.0 volts rms, maximum.

c. One voice channel with 300 to 3000 Hz bandwidth, 0.25 volts rms, maximum.

3.5.1.7.10 (U) Channel Phase Correlation - Difference in phase delay among sonic channels shall be less than 5 degrees, 10 to 500 Hz.

*3.5.1.7.11 (U) RF Channel Indicator - A means shall be employed to clearly depict the assigned channel letter by a movable, screw-secured tab.

*3.5.1.7.12 (U) Distortion - The baseband output shall contain no single harmonic of any of the four subcarrier signals or of the deviated subcarriers that is less than 40 db below any zero signal subcarrier amplitude.

*3.5.1.7.13 (U) Common Mode Rejection - The transmitter-multiplexer differential input shall provide 60 db common mode rejection at 1000 Hz.

*3.5.1.7.14 (U) Interchangeability with Similar Units in Other Systems - The acoustic voltage-controlled oscillators (paragraph 3.5.1.8), acoustic low-pass filters (paragraph 3.5.1.9), voice low-pass filter (paragraph 3.5.1.10), mixer amplifier (paragraph 3.5.1.11), isolation transformer assembly (paragraph 3.5.1.15.3), and the telemetry mount (paragraph 3.5.1.15) shall be one-way interchangeable with similar units in the TD-790/AKT-19.

*3.5.1.8 (U) Acoustic Voltage-Controlled Oscillators - This acoustic voltage-controlled oscillators shall meet the following requirements:

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*3.5.1.8.1 (U) Function - The four acoustic voltage-controlled oscillators shall be nonrepairable solid-state plug-in modules that convert filtered data signals, received from a corresponding low-pass filter module, into frequency-modulated sine wave restored subcarrier signals.

*3.5.1.8.2 (U) Input Voltage - Isolated 28 vdc from acoustic low-pass filter.

*3.5.1.8.3 (U) Input Signal Voltage (Modulating Voltage) - ±2.5 volts peak-to-peak at 0 to 2000 Hz for full-scale deviation.

3.5.1.8.4 (U) Operating Frequency Assignments and Deviation

Design Center Frequency	Maximum Deviation
12.5 kHz	±2 kHz
20.8 kHz	±2 kHz
45.8 kHz	±2 kHz
54.2 kHz	±2 kHz

3.5.1.8.5 (U) Harmonic Distortion - Less than 1 percent of any frequency in passband.

3.5.1.8.6 (U) Linearity - Deviation from best straight line shall be less than ±0.125% of design bandwidth.

3.5.1.8.7 (U) Amplitude Modulation - Less than 10 percent.

3.5.1.8.8 (U) Temperature Stability - Center frequency shall vary less than ±2.0% of design bandwidth.

*3.5.1.8.9 (U) Input Impedance - 400 kilohms minimum shunted by 5 pf, maximum for all channels.

*3.5.1.8.10 (U) Controls - The voltage-controlled oscillators shall have the following controls:

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AMPL control	Used to adjust amplitude.
SENS control (gain)	Used to adjust deviation bandwidth of voltage-controlled oscillator.
FREQ control (balance)	Used to set (zero) operating center frequency.

3.5.1.8.11 (U) Test Point - The voltage controlled oscillators shall have an output test point to monitor the fm subcarrier output signals.

3.5.1.8.15 (U) Form Factor - The dimensions of the voltage controlled oscillators shall be in accordance with figure 6.

*3.5.1.9 (U) Acoustic Low-Pass Filter (Sonic Data Channels) - The low-pass filters shall meet the following requirements:

*3.5.1.9.1 (U) Function - The four acoustic low-pass filters shall be depot repairable solid-state plug-in modules that provide signal conditioning for four sonic data channel inputs to multiplexer. They shall provide dc power, isolated from ground to corresponding acoustic voltage controlled oscillator modules.

3.5.1.9.2 (U) Input Power - 28 vdc supplied from internal regulated power of T-1220/AKT-22(V).

*3.5.1.9.3 (U) Input Signal Voltage - 16 volts rms, maximum.

*3.5.1.9.4 (U) Frequency Range (Bandpass) - The input signal shall be filtered to a bandpass of 10 to 2000 Hz. The passband shall be flat within ±0.5 db between 10 Hz and 500 Hz, rolling off to -3 ±1 db at 2 kHz. The signal shall be down 12 db minimum at 8 kHz.

*3.5.1.9.5 (U) Output Level - 1.77 volts rms at 100 Hz with 16 volt rms input.

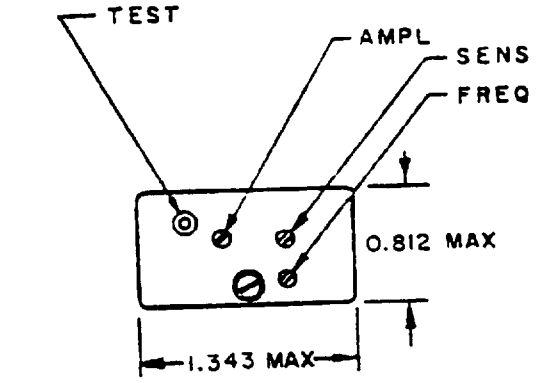
*3.5.1.9.6 (U) Controls - The LOFAR low-pass filter shall have an ADJUST control to adjust the signal level of the amplifier output circuit in the low-pass filter to within ±0.1 volt rms of specified output level.

*3.5.1.9.7 (U) Test Point - The low-pass filter shall have output and output return test points.

3.5.1.9.8 (U) Form Factor - The dimensions of the LOFAR low-pass filter shall be in accordance with figure 7.

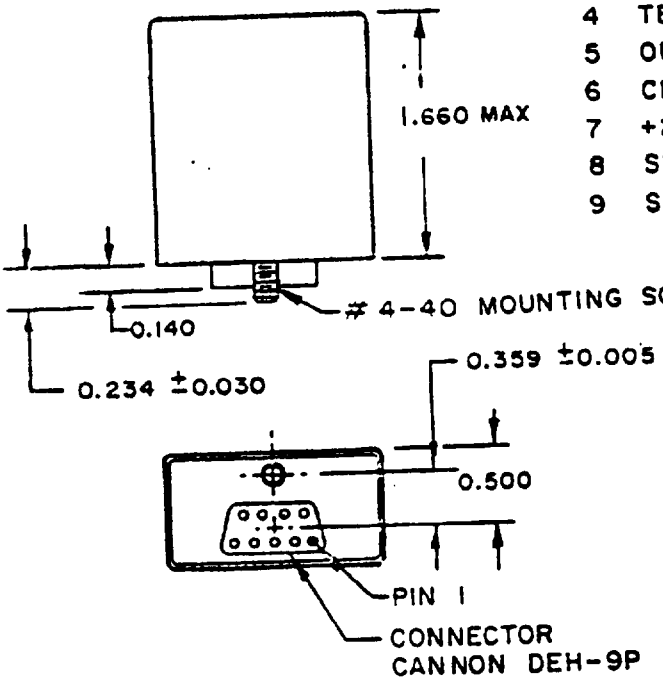
*3.5.1.9.9 (U) Output Voltage - 28 ±3 volts dc at 10 ma, isolated from power input ground.

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PIN FUNCTIONS:

1	INPUT
2	PWR AND SIG RETURN
3	SPARE
4	TEST OUTPUT
5	OUTPUT
6	CHASSIS GROUND
7	+28V DC
8	SPARE (INPUT NO. 2)
9	SPARE



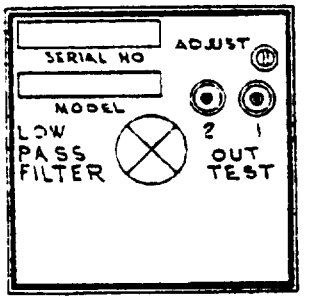
UNLESS OTHERWISE SPECIFIED

TOLERANCES

.XX ± .02
.XXX ± .005

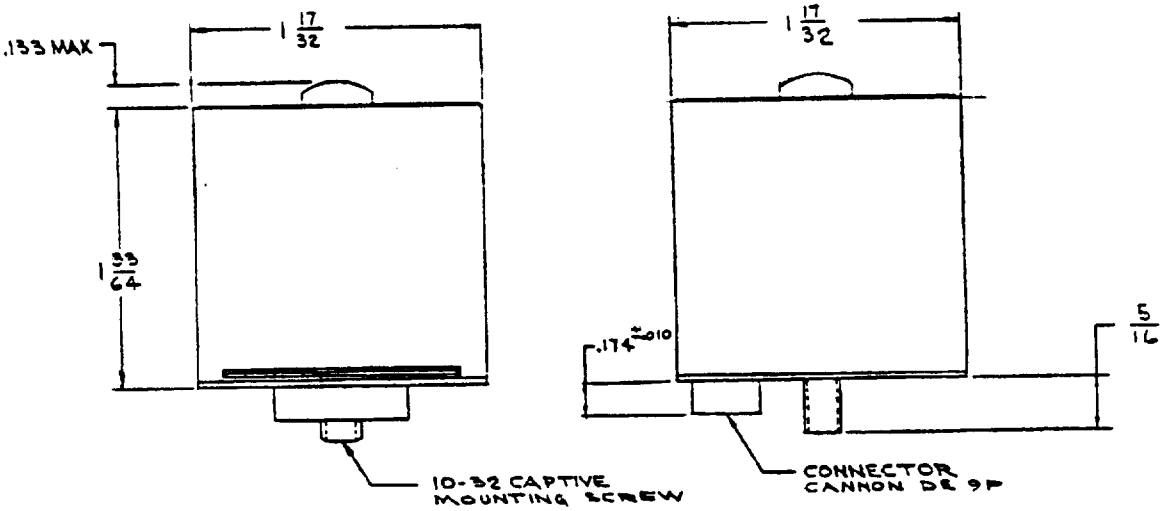
Figure 6. Voltage-Controlled Oscillators Form Factor and Connector Requirements

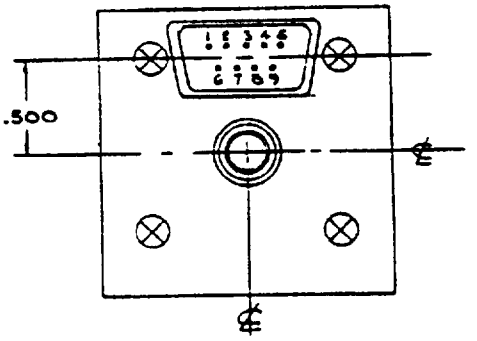
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.



Pin Functions

1. Diff Input (+)
2. Sig & 28 V Isolated Return
3. Diff Input (-).
4. Spare
5. Signal Out
6. Chassis Ground
7. +28 VDC Isolated
8. +28 VDC Input
9. 28 VDC Return





Note:

Tolerances unless noted:

Fractions:	±1/64
Decimals:	±0.005
Angles:	±1/2°

Figure 7. Low-Pass Filter, Form Factor and Connector Requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
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*3. 5. 1. 10 (U) Voice-Channel Low-Pass Filter - The voice-channel low-pass filter shall meet the requirements specified in paragraph 3. 5. 1. 9 with the following exceptions.

*3. 5. 1. 10. 1 (U) Function - The voice-channel low-pass filter shall be a depot repairable solid-state plug-in module that provides signal conditioning for voice-frequency channel inputs to the multiplexer. It shall provide dc power, isolated from ground to the mixer-amplifier.

*3. 5. 1. 10.2 (U) Input Signal Voltage - 250 millivolts rms, maximum.

*3. 5. 1. 10. 3 (U) Frequency Range (Bandpass) - The input shall be filtered to a bandpass of 300 Hz to 3 kHz. The passband shall be flat within ±1 db from 300 Hz to 2 kHz rolling off to -3 ±1 db at 3 kHz. The roll off shall be 6 db per octave, minimum beginning at 3 kHz.

*3. 5. 1. 10. 4 (U) Output Level - Adjustable to 80 millivolts rms for 250 millivolts rms input signal at 1000 Hz.

3. 5. 1. 11 (U) Mixer-Amplifier - The mixer-amplifier shall meet the following requirements:

*3. 5. 1. 11. 1 (U) Function - The mixer-amplifier shall be a solid-state depot repairable plug-in module that mixes the summed outputs of four voltage-controlled oscillators, voice-frequency low-pass filter and the trigger filter to provide an FM-multiplexed output signal on a common bus to the transmitter. The mixer-amplifier shall provide sufficient amplification and gain control to drive the transmitter through the isolation transformer assembly.

*3. 5. 1. 11. 2 (U) Input Power - Isolated 28 vdc from voice low-pass filter.

*3. 5. 1. 11. 3 (U) Input Signal Voltage - 2 volts rms, maximum.

*3. 5. 1. 11. 4 (U) Frequency Range (Bandpass) - 300 Hz to 250 kHz within ±0. 25 db.

*3. 5. 1. 11. 5 (U) Voltage Gain - 20 minimum, adjustable to 2. 5.

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*3.5.1.11.6 (U) Input Impedance - 100 kilohms, minimum shunted by 20 pf, maximum.

3.5.1.11.7 (U) Output Impedance - 50 ohms, maximum.

3.5.1.11.8 (U) Linearity - ±0.5% from best straight line.

*3.5.1.11.9 (U) Distortion - Less than 0.5%.

*3.5.1.11.10 (U) Controls - The mixer-amplifier shall have a GAIN control.

*3.5.1.11.11 (U) Test Points - The mixer-amplifier shall have input and output test points.

3.5.1.11.12 (U) Form Factor - The dimensions of the mixer-amplifier shall be in accordance with figure 8.

3.5.1.12 (U) Trigger-Filter - The trigger-filter shall meet the following requirements:

*3.5.1.12.1 (U) Function - The trigger filter shall be an IMA repairable plug-in module that shall provide passive frequency and noise filtering for up four AN/SSQ-47 sonobuoy trigger signals.

*3.5.1.12.2 (U) Passband - 26 to 38 kHz ±1.5 db.

*3.5.1.12.3 (U) Stopband - 20 db down; below 22.8 kHz; above 43.8 kHz.

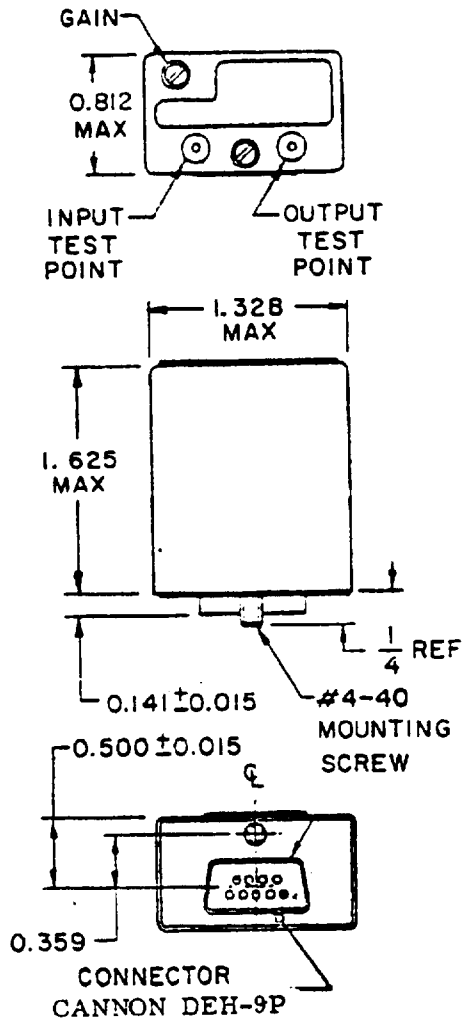
*3.5.1.12.4 (U) Input Impedance - 10 kilohm, nominal, each of four

*3.5.1.12.5 (U) Output Impedance - 10 kilohm, nominal.

3.5.1.12.6 (U) Input Signal Voltage - 1 ±0.1 volt rms.

*3.5.1.12.7 (U) Output Signal Voltage - Adjustable up to 50 millivolts rms with a single 1 volt rms input.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
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Connector Pin Functions:

PIN	FUNCTION
1	Input
2	Power & Signal Return
3	Spare
4	Spare
5	Output
6	Chassis Ground
7	−28v dc
8	Spare
9	Spare

NOTES:
1. ALL DIMENSIONS IN INCHES
2. TOLERANCES ±.005 EXCEPT AS NOTED.

Weight: 2.0 ounces (approximately).
Volume: 1.8 cubic inches.

Figure 8. Mixer Amplifier, Form Factor and Connector Requirements

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Check the source to verify that this is the current version before use.

3.5.1.12.8 (U) Form Factor - The dimensions of the trigger-filter shall be in accordance with figure 9.

*3.5.1.12.9 (U) Controls - The trigger filter shall have a gain control to provide adjustment 0 to 50 mv with 1 volt input.

3.5.1.13 (U) FM Transmitter - The FM transmitter shall meet the following requirements:

*3.5.1.13.1 (C) Function - The transmitter shall be a factory repairable encapsulated, solid-state, crystal-controlled fm transmitter module. It shall fm modulate and transmit multiplexed output signals on one of three standard assigned carrier frequencies. The transmitter shall be true fm and adjustable to accept external video, PCM waveform, or fm/fm multiplex.

*3.5.1.13.2 (U) Input Power - 28 ±1.0 volts dc at 2.5 amps, maximum as supplied from internal regulated power.

*3.5.1.13.3 (U) Input Impedance - 50 kilohm, minimum shunted by 30 pf, maximum.

*3.5.1.13.4 (C) Output Power - 5.5 watts, minimum, 8.0 watts maximum.

*3.5.1.13.5 (U) Output Load Impedance - 50 ohms, VSWR 1.3:1, maximum.

*3.5.1.13.6 (U) Frequency Range - 2200 to 2290 MHz.

*3.5.1.13.7 (U) Frequency Stability - 0.003 percent of center frequency; ±0.005 over environments.

*3.5.1.13.8 (C) Frequency Deviation Sensitivity - 2.8 volts peak-to-peak for ±125 kHz ±10 percent.

*3.5.1.13.9 (C) Total Harmonic Distortion - Less than 2 percent at 125 kHz deviation.

*3.5.1.13.10 (C) Deviation Linearity - ±2 percent best straight line.

*3.5.1.13.11 (U) Incidental FM - Less than ±5 kHz over modulation range.

3.5.1.13.12 (U) Modulation Type - True fm.

3.5.1.13.13 (U) Form Factor - The dimensions of the transmitter shall be in accordance with figure 10.

*3.5.1.13.14 (C) Modulation Range - 10 Hz to 125 kHz, flat within ±2 db.

*3.5.1.14 (U) Isolator/Power Monitor - The isolator/power monitor shall meet the following requirements.

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FUNCTION TABLE

Pin	Function
J1-1	Input 5 Signal
J1-3	Input 5 Return
J2-1	Input 6 Signal
J2-3	Input 6 Return
J3-1	Input 7 Signal
J3-3	Input 7 Return
J4-1	Input 8 Signal
J4-3	Input 8 Return
J5-2	Output Return
J5-5	Output Signal

NOTES: 1. Connectors J1 through J5 are type 14073-9P.

2. Tolerances unless noted:

Fractions: ±1/64
Decimals: .xxx±0.005
.xx±0.01

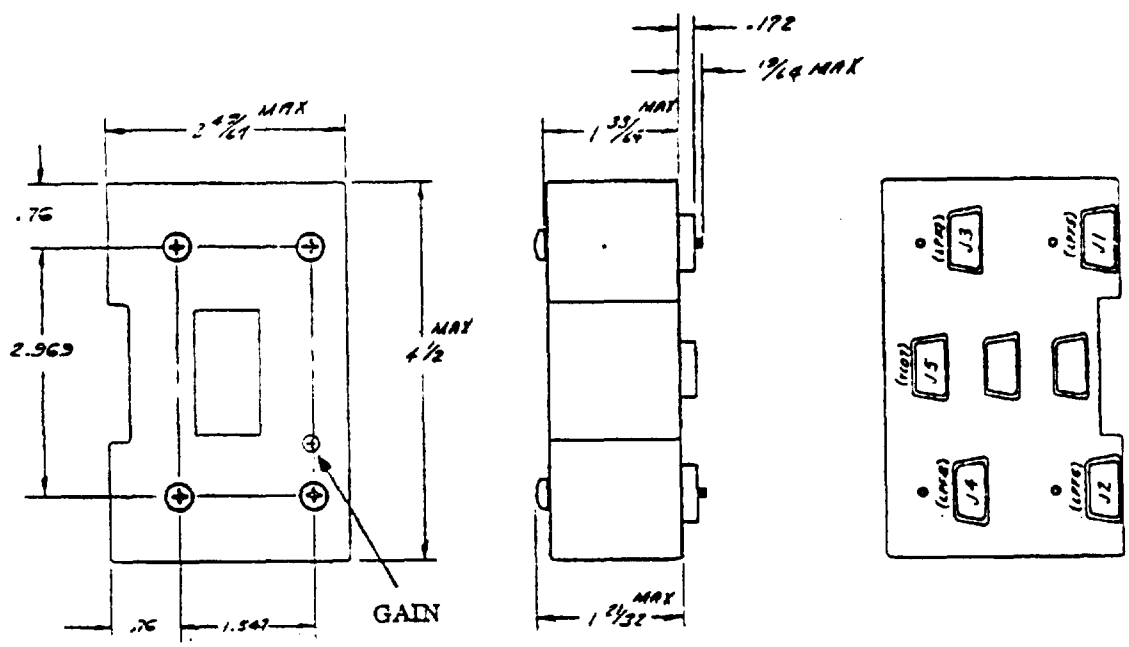
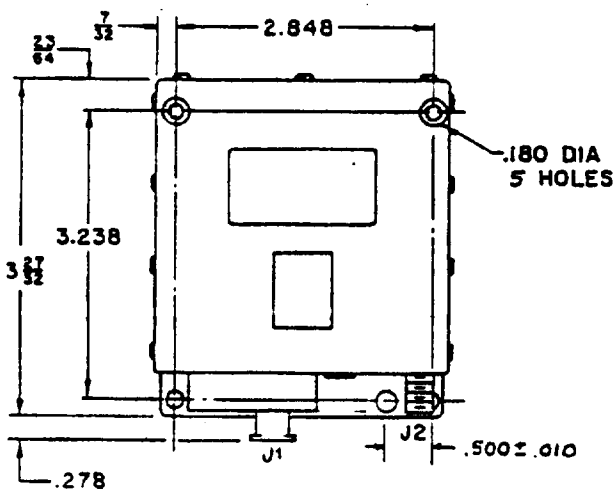
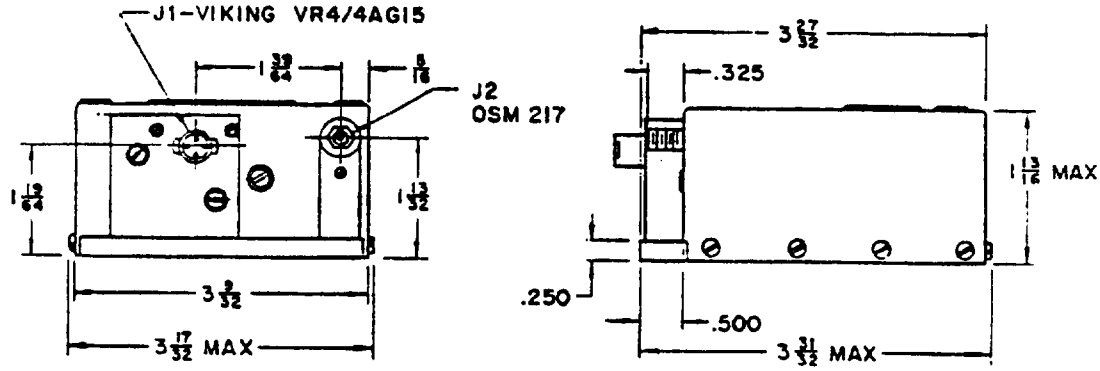


Figure 9. Trigger Filter, Form Factor and Connector Requirements

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CONNECTOR PIN	PIN FUNCTIONS FUNCTION
J1	INPUT
A	+28VDC
B	GROUND
C	MOD
D	GROUND
J2	OUTPUT RF



NOTE:

TOLERANCES UNLESS NOTED:
FRACTIONS: ±1/64
DECIMALS: ±0.005
ANGLES: ±1/2°

Figure 10. FM Transmitter, Form Factor and Connector Requirements

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*3.5.1.14.1 (U) Function - The isolator/power monitor shall be a non-repairable encapsulated unit that provides transmitter output and external load isolation, and shall have integral forward and reverse power monitoring capabilities.

*3.5.1.14.2 (C) RF Power Handling Capability - 10 watts, minimum.

3.5.1.14.3 (U) Reverse Isolation - 20 db, minimum.

3.5.1.14.4 (C) Frequency Range (Passband) - 2200 to 2300 MHz.

3.5.1.14.5 (U) Termination Impedance - 50 ohms.

3.5.1.14.6 (U) Voltage Standing Wave Ratio - 1.25:1, maximum. .

3.5.1.14.7 (U) Forward Insertion Loss - 0.3 db, maximum.

*3.5.1.14.8 (C) Power Monitoring Outputs - 25 ±2.5 microamperes at 5 watts forward and reverse, into 50 microampere ammeter with 20-kilohm impedance.

3.5.1.14.9 (U) Form Factor - The dimensions of the isolator shall be in accordance with figure 11.

*3.5.1.15 (U) Telemetry Mount - The telemetry mount shall meet the following requirements.

*3.5.1.15.1 (U) Function - The telemetry mount shall be an IMA repairable assembly that provides mounting facilities and interconnecting wiring for plug-in modules and transformer assembly. It shall be wired and have provisions for nine low-pass filters, eight voltage controlled oscillators, two DIFAR modules, and a wideband mixer-amplifier. The isolation transformer assembly shall be mounted and wired below the chassis. Signal, power and control connections shall be made through connectors J21 and J22. The provisions for the two DIFAR modules shall consist of two wired and unused sockets (J19 and J20). As used in the (V)1 unit without the DIFAR multiplex, a jumper shall be installed between pins 5 and 8 of J20. When the DIFAR modules are used, as in the (V)2 unit, the jumper shall not be installed.

*3.5.1.15.2 (U) Form Factor - The dimensions of the telemetry mount shall be in accordance with figure 12.

3.5.1.15.3 (U) Isolation Transformer Assembly - The isolation transformer assembly in the multiplex mount shall meet the following requirements.

*3.5.1.15.3.1 (U) Function - The isolation transformer assembly shall be a nonrepairable encapsulated chassis-mounted assembly that contains the isolation transformer and attenuation network used to couple multiplexed output to fm transmitter.

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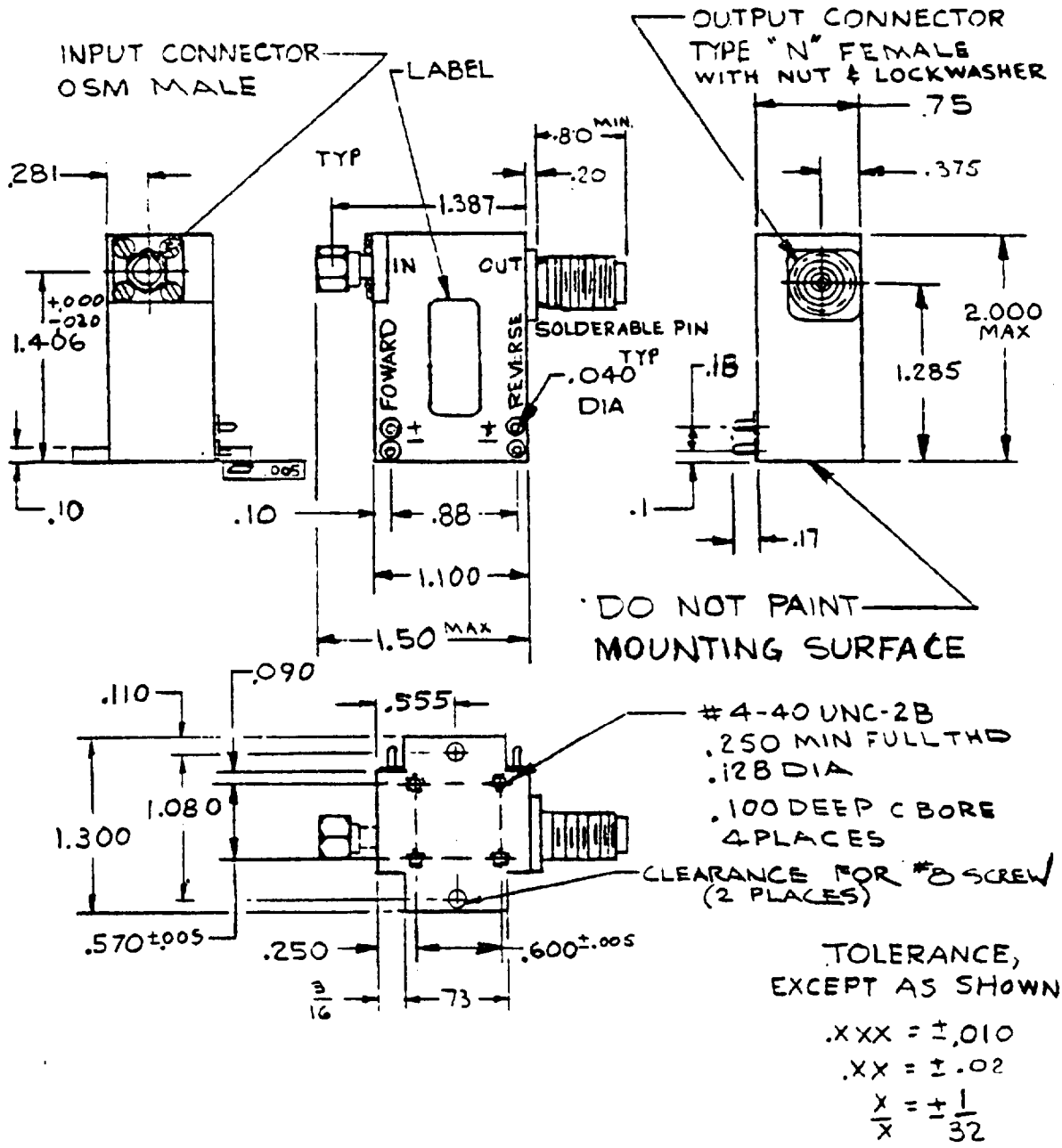


Figure 11. Isolator/Power Monitor Form Factor and Con

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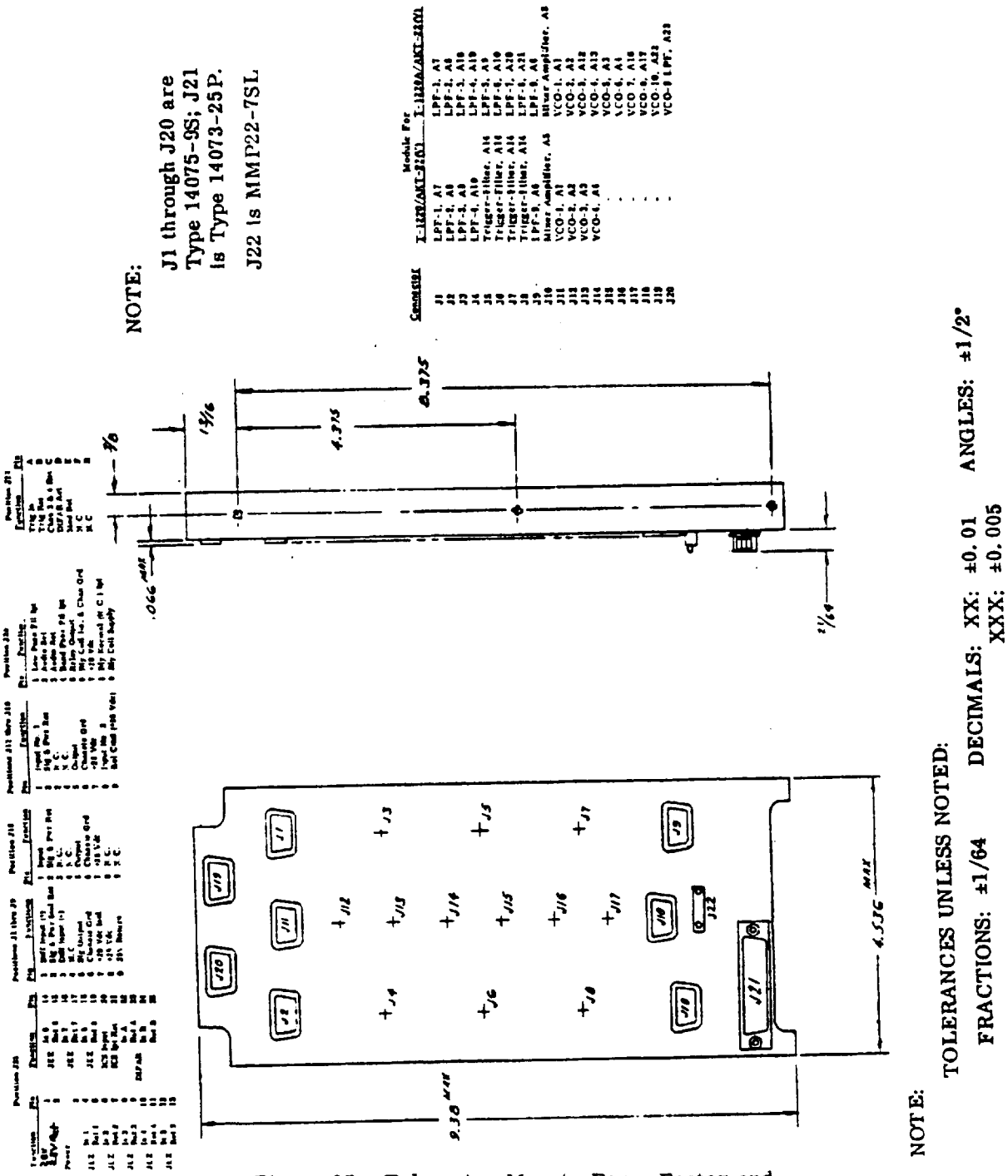


Figure 12. Telemetry Mount, Form Factor and Connector Requirements

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3.5.1.15.3.2 (U) Transformer Ratio - 1 to 2 voltage stepup with a 2.5 kilohm primary and 10 kilohm secondary.

3.5.1.15.3.3 (U) Output Attenuation - 2 to 1 resistive to produce a nominal unity gain overall circuit.

3.5.1.15.3.4 (U) Input Voltage - 1 volt rms maximum.

*3.5.1.15.3.5 (U) Common Mode Rejection - 10 megohm, minimum at dc and 40 db, minimum at 400 Hz.

*3.5.1.15.3.6 (U) Response - ±3 db from 250 Hz to 3000 kHz.

3.5.1.15.3.7 (U) Form Factor - The dimensions of the isolation transformer assembly shall be in accordance with figure 13.

*3.5.2 (U) Transmitter-Multiplexer T-1220A/AKT-22(V) - The transmitter-multiplexer shall meet the following requirements.

*3.5.2.1 (U) Function - The T-1220A/AKT-22(V) transmitter-multiplexer shall condition and combine one voice channel and up to eight sonic channels; one voice channel, up to six sonic channels, and up to four trigger signals; or two DIFAR channels into a frequency-division mutliplex signal. It shall fm modulate a uhf carrier with the combined multiplex signal and amplify the carrier. The T-1220A/AKT-22(V) shall be convertible to completely or partially perform the functions of the T-1220/AKT-22(V) by the deletion, transfer, or replacement of modules. Provision for modulation by an external video or PCM signal shall be provided. The T-1220A/AKT-22(V) shall be LMA repairable.

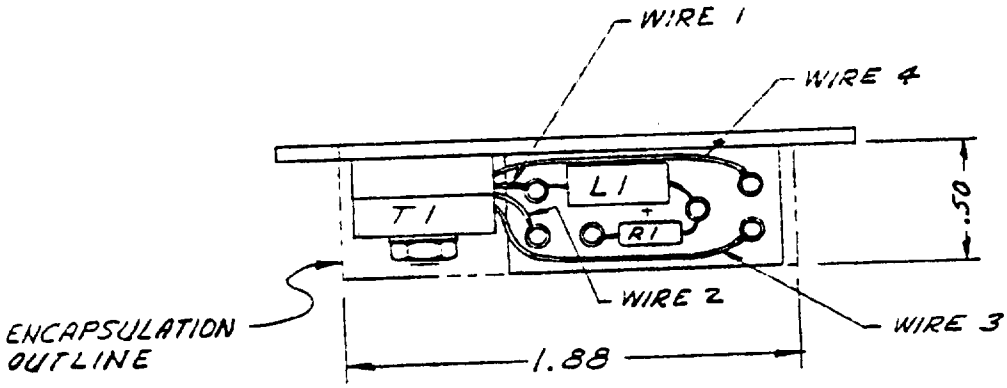
*3.5.2.2 (U) Form Factor - The transmitter-multiplexer shall be enclosed in a case with dimensions as defined in figure 5. The connector pin designations shall be as specified in figure 14.

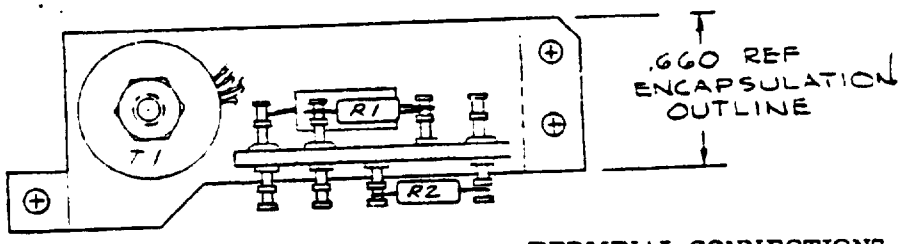
*3.5.2.3 (U) Weight - The weight of the T-1220A/AKT-22(V) shall not exceed 21 pounds.

*3.5.2.4 (U) Contents - The transmitter-multiplexer shall contain the following major assemblies:

- Seven acoustic voltage controlled oscillators per paragraph 3.5.2.8.
- One wideband acoustic voltage controlled oscillator per paragraph 3.5.2.8.
- One DIFAR voltage controlled oscillator per paragraph 3.5.2.9.
- Seven acoustic low-pass filters per paragraph 3.5.2.10.
- One wideband low-pass filter per paragraph 3.5.2.10.
- One voice-channel low-pass filter same as for the T-1220/AKT-22(V) per paragraph 3.5.1.10.
- One mixer-amplifier same as for the T-1220/AKT-22(V) per paragraph 3.5.1.11.

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NOTE:

TOLERANCES:
DECIMALS: .XX ±0.02

TERMINAL CONNECTIONS:

E1 - Signal Input
E2 - Isolated Power and Signal Return
E3 - Signal Output
E4 - Output Return (Gnd)

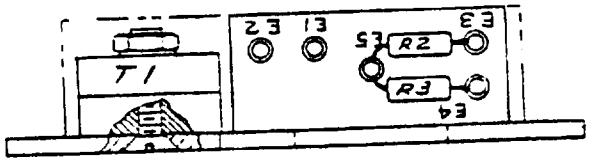


Figure 13. Isolation Transformer Assembly, Form Factor

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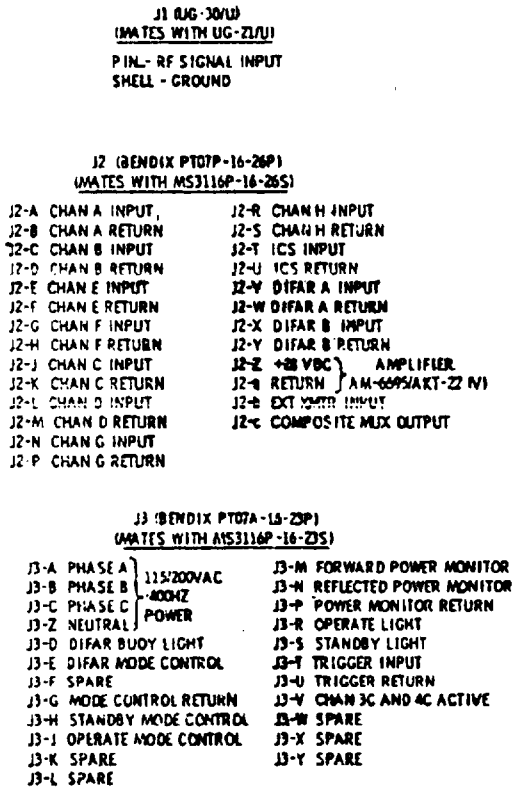


Figure 14. Transmitter-Multiplexer T-1220A/AKT-22 Connector Pin Designations

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- One DIFAR multiplex filter per paragraph 3.5.2.11.
- One fm transmitter same as for the T-1220/AKT-22(V) per paragraph 3.5.1.13.
- One rf power amplifier per paragraph 3.5.2.12.
- One isolator/power monitor per paragraph 3.5.2.13.
- One telemetry mount per paragraph 3.5.2.14.
- One power supply with control circuitry

*3.5.2.5 (U) Controls - The only operating control on the front panel of the transmitter-multiplexer shall be a circuit breaker for equipment protection.

*3.5.2.6 (U) Electrical Connections - Connections to external circuits shall be as provided for the T-1220/AKT-22(V) as specified in paragraph 3.5.1.6.

*3.5.2.7 (U) Overall Unit - The overall transmitter-multiplexer shall meet the requirements for the T-1220/AKT-22(V) as specified in paragraph 3.5.1.7 with the following exceptions:

*3.5.2.7.1 (U) Operating Modes -

a. STANDBY mode, shall turn on and apply power to multiplexer circuits.

b. OPERATE mode, shall apply operational power to transmitter, and shall enable multiplexing and transmitting of one voice channel and active/passive data in one of the following two modes:

(1) With no triggers enabled for AN/SSQ-47 sonobuoys, eight sonobuoy signals from any mix of AN/SSQ-41, AN/SSQ-36, or the LOFAR component of AN/SSQ-53 sonobuoys shall be multiplexed and transmitted.

(2) With up to four triggers enabled for AN/SSQ-47 sonobuoys, six sonobuoy signals from a mix of up to four AN/SSQ-47 and the other signals from any mix of AN/SSQ-41, AN/SSQ-36, or the LOFAR component of AN/SSQ-53 sonobuoys shall be multiplexed and transmitted.

c. DIFAR mode, shall enable multiplexing and transmitting of two composite DIFAR signals.

d. OFF mode, no power required.

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Check the source to verify that this is the current version before use.

*3.5.2.7.2 (C) Operating Frequency - One of the following as indicated by letter or color on transmitter and transmitter-multiplexer:

A - Red - 2212.5 MHz

B - Green - 2220.5 MHz

C - Yellow - 2240.5 MHz

D - White - 2256.5 MHz

E - Black - 2276.5 MHz

*3.5.2.7.3 (C) Deviations - The deviations to the main carrier for the various subcarrier and baseband signals are:

Subcarrier (kHz)	Deviation (kHz)
12.5	±14.3
20.8	±14.3
29.2	±14.3
37.5	±14.3
45.8	±18.7
54.2	±21.8
62.5	±25.0
93.0	±37.0
Voice (Baseband)	± 9.4
Triggers (Each)	± 6.2
DIFAR A	±100.0
DIFAR B	±50.0

*3.5.2.7.4 (C) RF Power Output - 12.6 watts +2, -1 db into 50-ohm resistive load over entire frequency range from -54°C to +55°C.

*3.5.2.7.5 (U) Signal Inputs -

a. OPERATE Mode - with no trigger switches on -

1. Seven passive sonic data channels with 10 to 2000 Hz bandwidth, 16 volt rms for full VCO deviation.
2. One passive sonic data channel with 10 to 2800 Hz bandwidth, 16 volt rms, for full VCO deviation.
3. One voice channel with 300 to 3000 Hz bandwidth, 0.25 volt rms, maximum.

b. OPERATE Mode - with one or more trigger switches on -

1. Five passive sonic data channels with 10 to 2000 Hz bandwidth, 16 volt rms for full VCO deviation. Up to four may be used for range only echoes.
2. One passive sonic data channel with 10 to 2800 Hz bandwidth, 16 volt rms for full VCO deviation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
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3. Up to four range only sonar triggers in 26 to 38 kHz bandwidth,1 volt rms, maximum.

4. One voice channel with 300 to 3000 Hz bandwidth, 0.25 volt rms, maximum.

c. DIFAR Mode

1. Two composite DIFAR channels 10 Hz to 17.4 kHz, up to 10 volts peak-to-peak.

*3.5.2.7.6 (U) Channel Phase Correlation - Difference in phase delay between any two sonic channels shall be less than 5 degrees, 10 to 500 Hz, in seven 2-kHz bandwidth LOFAR channels.

*3.5.2.8 (U) Acoustic Voltage-Controlled Oscillators - The eight acoustic voltage-controlled oscillators shall meet the requirements specified in paragraph 3.5.1.8 for the T-1220/AKT-22(V) with the following additional requirements.

*3.5.2.8.1 (U) Input Signal Voltage (Modulating Voltage) - ±2.5 volts peak-to-peak at 10 to 2000 Hz for full scale deviation for all channels except 93.0 kHz: ±2.5 volts peak-to-peak at 10 to 2800 Hz for full scale deviation for the 93.0 kHz channel (wideband) VCO.

3.5.2.8.2 (U) Operating Frequency Assignments and Deviations -

Design Center Frequency	Maximum Deviation
29.2 kHz	±2 kHz
37.5 kHz	±2 kHz
62.5 kHz	±2 kHz
93.0 kHz	±13.95 kHz

*3.5.2.9 (U) DIFAR Voltage Controlled Oscillator - The DIFAR voltage controlled oscillator shall meet the following requirements:

*3.5.2.9.1 (U) Function - The voltage-controlled oscillator shall be a depot repairable solid-state plug-in module that provides a frequency-modulated subcarrier (70 kHz) for DIFAR channel B signals.

*3.5.2.9.2 (U) Input Power - 28 vdc supplied from internal regulated power of T-1220A/AKT-22(V).

*3.5.2.9.3 (U) Input Signal Voltage - Up to 10 volt peak-to-peak maximum, offset from 0V; response of 10 Hz to 17.4 kHz.

*3.5.2.9.4 (U) Subcarrier Center Frequency and Deviation - 70 kHz ±40%

3.5.2.9.5 (U) Modulation Sensitivity - 10 volt, peak-to-peak for full-scale deviation

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*3.5.2.9.6 (U) Output Harmonic Distortion - Quasi square wave.

3.5.2.9.7 (U) Linearity - Deviation from best straight line shall be less than ±1.0% design bandwidth.

3.5.2.9.8 (U) Amplitude Modulation - Less than 10 percent.

*3.5.2.9.9 (U) Temperature Stability - Center frequency shall vary less than ±5.0 percent of design bandwidth.

3.5.2.9.10 (U) Output Voltage - Adjustable up to 3 vac peak-to-peak into 10 kilohm load.

*3.5.2.9.11 (U) Form Factor - The dimensions of the DIFAR voltage controlled oscillator shall be in accordance with figure 6.

*3.5.2.9.12 (U) Controls - The DIFAR voltage controlled oscillators shall have the following controls.

AMPL control	Used to adjust amplitude.
SENS control	Used to adjust deviation bandwidth of voltage controlled oscillator.
FREQ control	Used to set (zero) operating center frequency.

*3.5.2.9.13 (U) Test Point - The DIFAR voltage controlled oscillators shall have an output test point to monitor the fm subcarrier output signals.

*3.5.2.10 (U) LOFAR Low-Pass Filters (Sonic Data Channels) - The eight low-pass filters shall meet the requirements specified in paragraph 3.5.1.9 for the T-1220/AKT-22(V) with the following additional requirement for the wideband low-pass filter.

*3.5.2.10.1 (U) Frequency Range (Bandpass) - The wideband input signal shall be filtered to a bandpass of 10 to 2800 Hz. The passband shall be flat within ±0.5 db between 10 Hz and 500 Hz rolling off to -3 ±1 db at 2.8 kHz. The signal shall be down 12 db, minimum at 8 kHz.

*3.5.2.11 (U) DIFAR Filter - The DIFAR filter shall meet the following requirements:

3.5.2.11.1 (U) Function - The filter shall be a plug-in module that performs two functions in the multiplexer section: passive filtering and linear combination of DIFAR channel inputs, and multiplexer output selection. The module shall contain a linear low-pass filter, a bandpass filter, a mixing network, and a relay.

3.5.2.11.2 (U) Input Control -+24 to 29 vdc.

3.5.2.11.3 (U) Baseband Signal Voltage - Up to 10 ±1 volt peak-to-peak.

3.5.2.11.4 (U) Subcarrier Voltage - Up to 3 volts peak-to-peak square wave.

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*3.5.2.11.5 (U) Passband - Low pass filter - 10 Hz to 17.4 kHz
Bandpass filter - 42 kHz to 98 kHz

3.5.2.11.6 (U) Rejection - Low-pass filter -

-20 db min at 42 kHz
-36 db min at 70 kHz

Bandpass filter -

-20 db min at 18 kHz
-30 db min at 150 kHz

3.5.2.11.7 (U) Load Impedance - 100 kilohms nominal.

3.5.2.11.8 (U) Output Impedance - 20 kilohms nominal.

*3.5.2.11.9 (U) Output Level - 3 volts, peak-to-peak nominal, adjustable ±25% with a ratio of two parts baseband to one part subcarrier.

*3.5.2.11.10 (U) Controls - The DIFAR filter shall have the following controls:

LEVEL Control	Used to adjust output level.
RATIO Control	Used to adjust ratio of baseband (DIFAR A) and subcarrier (DIFAR B) signals.

*3.5.2.11.11 (U) Test Point - The DIFAR filter shall have an output TEST point.

3.5.2.11.12 (U) Form Factor - The dimensions of the filter shall be in accordance with figure 15.

3.5.2.12 (U) Power Amplifier - The power amplifier shall meet the following requirements:

3.5.2.12.1 (U) Function - The power amplifier shall provide additional rf amplification of the transmitter output signal.

3.5.2.12.2 (U) Operating Voltage - Unregulated 35 vdc at 2.7 amps max.
Regulated 28 ±0.5 vdc at 0.27 amps max.

3.5.2.12.3 (U) Input Impedance - 50 ohms nominal.

3.5.2.12.4 (U) Input/Output Frequency Range - 2200 to 2290 MHz.

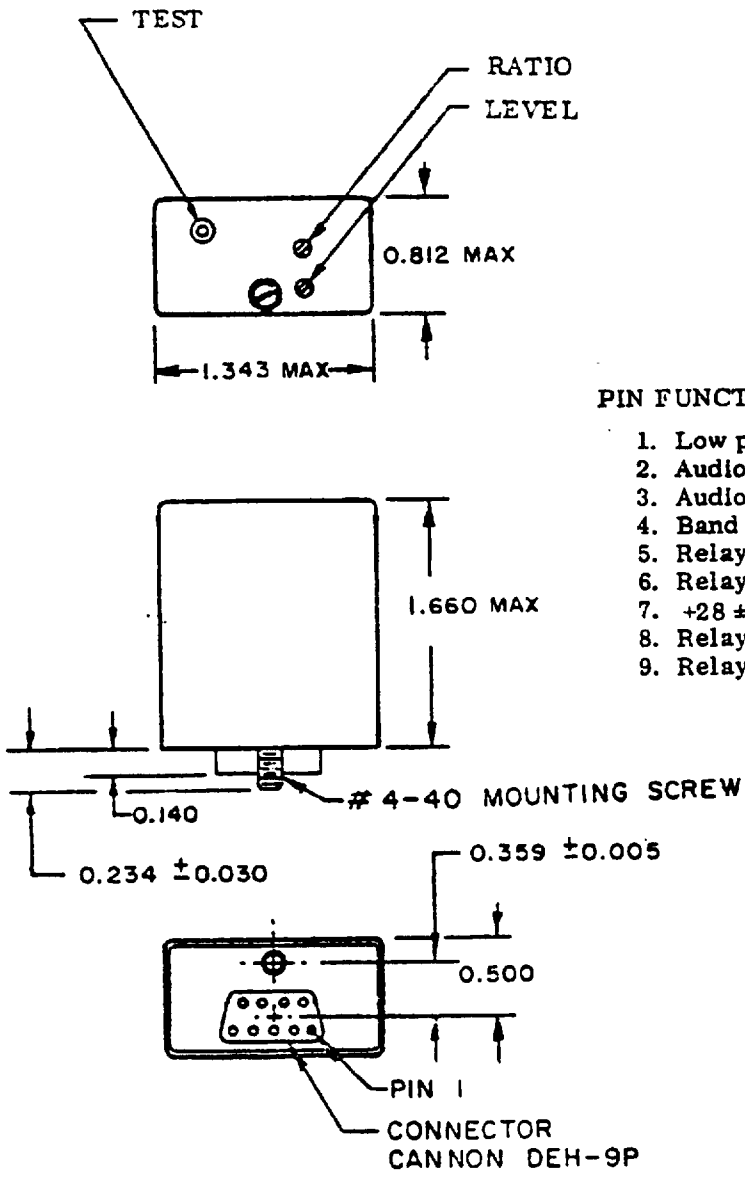
3.5.2.12.5 (U) Input/Output VSWR - 1.5:1 maximum.

*3.5.2.12.6 (C) Input RF Power - 5.0 to 8.0 watts.

*3.5.2.12.7 (C) Output Power - 11.0 to 20 watts.

*3.5.2.12.8 (U) Form Factor - The dimensions of the power amplifier shall be in accordance with figure 16.

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PIN FUNCTIONS

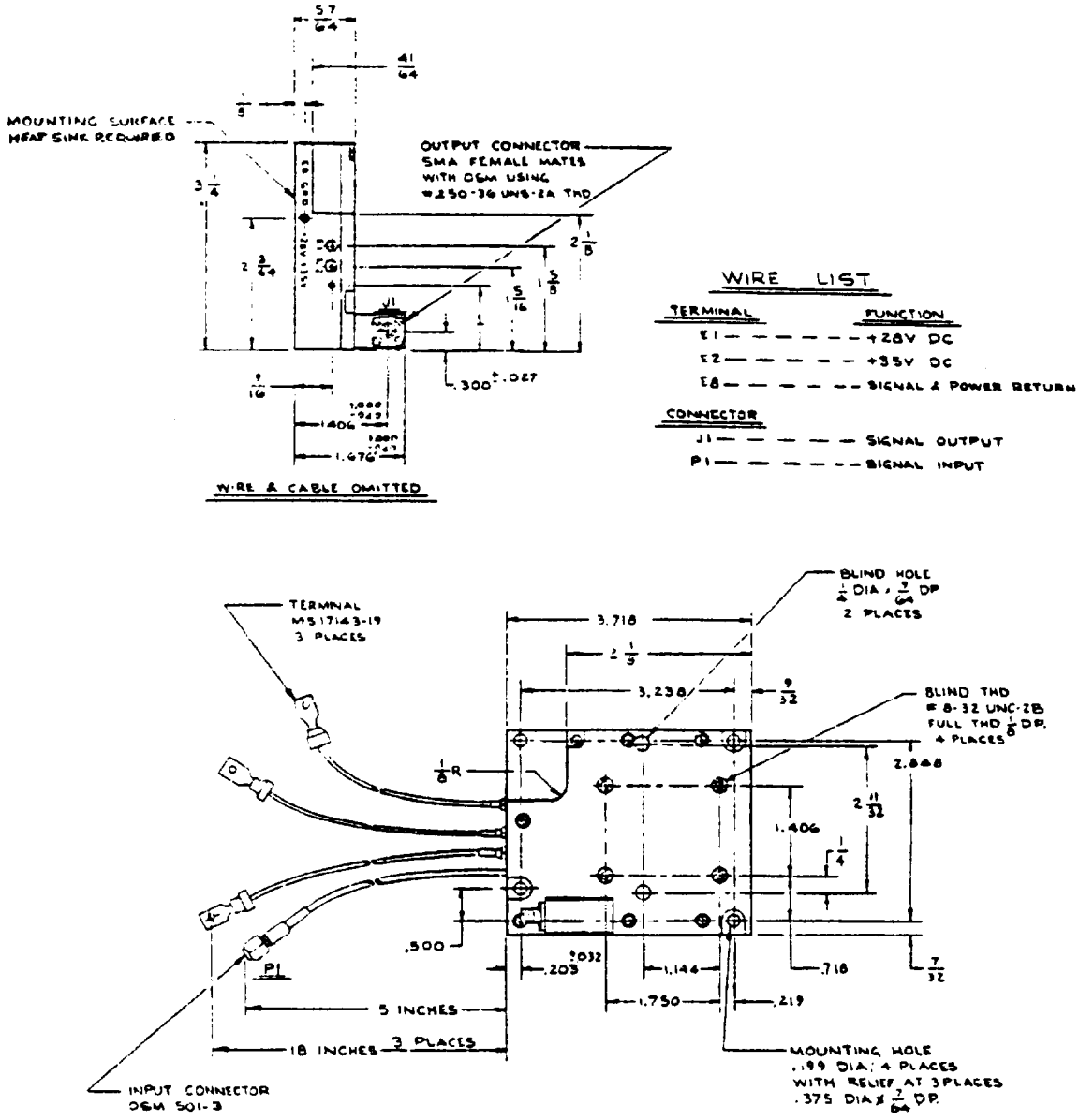
1. Low pass filter input
2. Audio return
3. Audio return
4. Band pass filter input
5. Relay output
6. Relay coil return & chassis ground
7. +28 ±4 vdc
8. Relay normal (N. C.) input
9. Relay coil supply

UNLESS OTHERWISE SPECIFIED

TOLERANCES
.XX ±.02
.XXX ±.005

Figure 15. DIFAR Filter, Form Factor and Connector Requirements

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NOTE: Tolerances unless otherwise specified:

Fractions: ±1/64
Decimals: ±0.005
Angles: ±1/2°

Figure 16. Power Amplifier Form Factor and Connector Requirements

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*3.5.2.13 (U) Isolator/Power Monitor - The isolator/power monitor shall meet the requirements specified in paragraph 3.5.1.14 for the T-1220/AKT-22(V) with the following exception:

*3.5.2.13.1 (C) RF Power Handling Capability - 20 watts, minimum.

*3.5.2.14 (U) Telemetry Mount - The multiplex mount shall meet the requirements specified in paragraph 3.5.1.15 with the following exceptions:

*3.5.2.14.1 (U) Function - The function of the telemetry mount for the T-1220A/AKT-22(V) shall be similar to that for the T-1220/AKT-22(V). The jumper in the mount between pins 3 and 8 of J20 for the T-1220/AKT-22(V) shall not be in the mount for the T-1220A/AKT-22(V). Also, an additional isolator resistor shall be installed between connector J21, pin 22 and connector J20, pin 1. The other differences shall be the complement of modules that are installed.

*3.5.3 (U) Transmitter-Multiplexer T-1295/AKT-22(V) - The transmitter-multiplexer shall meet the following requirements:

*3.5.3.1 (U) Function - The T-1295/AKT-22(V) transmitter-multiplexer shall condition and combine four sonic data channels, one voice channel, and up to four trigger signals, into a frequency-division multiplex signal and fm modulate a uhf carrier with the combined multiplex signal and amplify the carrier. The T-1295/AKT-22(V) shall perform the functions of the T-1220/AKT-22(V) with increased rf power and channel capability. The T-1295/AKT-22(V) shall be convertable to completely or partially perform the functions of the T-1220A/AKT-22(V) or T-1220B/AKT-22(V) by the addition, replacement, or transfer of modules. Provision for modulation by an external video or PCM signals shall be provided. The transmitter-multiplexer shall be IMA repairable.

*3.5.3.2 (U) Form Factor - The transmitter-multiplexer shall be enclosed in a case with dimensions shown in figure 17.

*3.5.3.3 (U) Weight - The weight of the transmitter-multiplexer shall not exceed 19 pounds.

*3.5.3.4 (U) Contents - The transmitter-multiplexer shall contain the following major assemblies:

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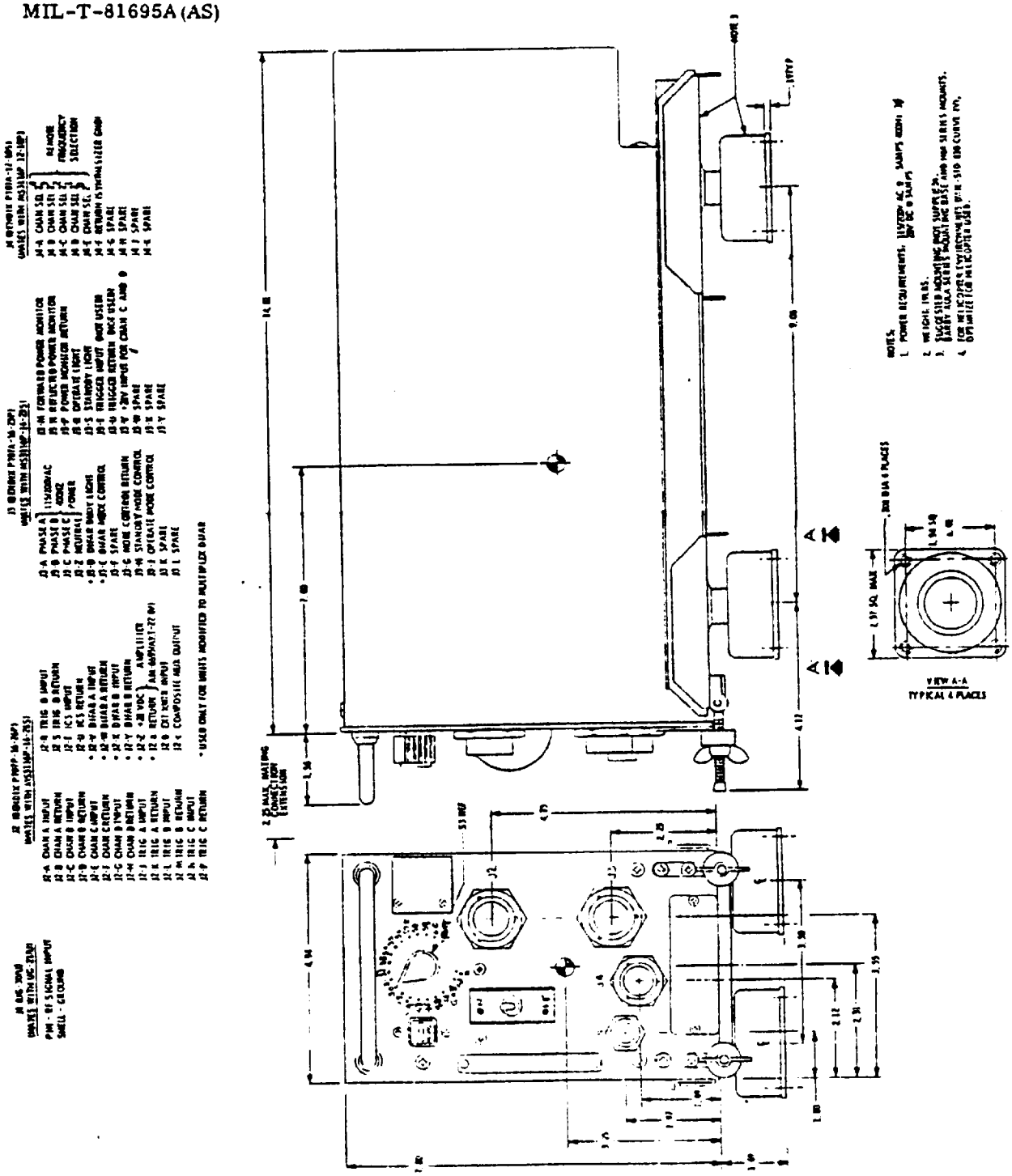


Figure 17. Transmitter-Multiplexer T-1295/AKT-22(V) Form Factor and Connector Requirements

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- Four acoustic voltage controlled oscillators same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 8.
- Four acoustic low-pass filters same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 9.
- One voice low-pass filter same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 10.
- One mixer-amplifier same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 11.
- One trigger filter same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 12.
- One 20-channel fm transmitter per paragraph 3. 5. 3. 8.
- One interstage isolator per paragraph 3. 5. 3. 9.
- One power amplifier per paragraph 3. 5. 3. 10.
- One isolator/power monitor per paragraph 3. 5. 3. 11.
- One telemetry mount same as for the T-1220/AKT-22(V) per paragraph 3. 5. 1. 15.
- One power supply and necessary control circuitry.

*3. 5. 3. 5 (U) Controls - The T-1295/AKT-22(V) shall have the following controls on the front panel:

Circuit Breaker Switch	Applies ac power to unit. Protects unit in the event of overload.
Channel Selector Switch	Provides 5-digit code (ground or open circuit) to transmitter to control transmit frequency for channels 41 through 60. In RMT position allows control of channel from remote switch.
DIFAR Input Switch	Behind front panel - wired only.

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*3.5.3.6 (U) Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	UG30( )/U (Type N)	RF Output
J2	PT07P-16-26P	Signal Inputs
J3	PT07A-16-23P	Power and Control
J4	PT07A-12-10S	Remote RF Channel Selection

*3.5.3.7 (U) Overall Unit - The T-1295/AKT-22(V) shall meet the requirements for the T-1220/AKT-22(V) as specified in paragraph 3.5.1.7 with the following exceptions:

*3.5.3.7.1 (C) Operating Frequency - Any one of the following rf channels may be selected on the front panel selection switch or remotely if so configured:

Channel		Frequency (MHz)
41	Red	2212.5
42		2216.5
43	Green	2220.5
44		2224.5
45		2228.5
46		2232.5
47		2236.5
48	Yellow	2240.5
49		2244.5
50		2248.5
51		2252.5
52	White	2256.5
53		2260.5
54		2264.5
55		2268.5
56		2272.5
57	Black	2276.5
58		2280.5
59		2284.5
60		2288.5

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*3.5.3.7.2 (C) RF Power Output - 25 watts +2, -1 db into 50-ohm resistive load over frequency range of 2200 to 2290 MHz from -54°C to 55°C.

*3.5.3.8 (U) FM Transmitter, 20 Channel - The 20-channel FM transmitter shall meet the following requirements:

*3.5.3.8.1 (U) Function - The 20-channel FM transmitter shall be a factory repairable encapsulated, solid-state, crystal controlled unit comprised of two separable units; an rf modulator and a frequency synthesizer.

*3.5.3.8.2 (C) Overall - The transmitter shall be a solid state, twenty channel selectable transmitter operating at S-band between 2200 and 2290 MHz. Channel selection shall be accomplishing by a remote control panel with no more than seven wires connecting control unit to transmitter. The transmitter shall be capable of synthesizing any of the ninety assigned IRIG channels at S-band. Selection of twenty of these IRIG channels shall be made by the insertion of a Read Only Memory (ROM) in the transmitter. The frequency synthesizer shall produce a reference signal to which rf modulator oscillator, operating at the final output frequency, can be phase locked.

The transmitter design shall be based on a phase-locked loop with the rf carrier producing oscillator operating at the output frequency. The oscillator output shall be amplified to the required output level.

A direct 70-MHz input interface shall be provided. This interface shall permit direct application of an angular modulated signal which is transferred to the S-band output. The interface shall be compatible with MIL-STD-188C levels with an input impedance of 75 ohms and an input level of +1 dbm.

*3.5.3.8.2.1 (U) Input Power - +28 ±4 volts dc at 2.7 amperes maximum. Internal polarity protection shall be provided.

*3.5.3.8.2.2 (U) Modulation - True FM.

*3.5.3.8.2.3 (U) Modulation Response - DC to 1 MHz ±1.0 db.

*3.5.3.8.2.4 (U) Modulation Input Impedance - 10,000 ohm resistive minimum, shunted by 30 pf maximum.

*3.5.3.8.2.5 (U) Deviation Sensitivity - ±125 kHz for 2.3 volts peak-to-peak ±10%.

*3.5.3.8.2.6 (U) Carrier Deviation - ±1 MHz maximum.

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*3. 5. 3. 8. 2. 7 (U) Linearity - ±1% of best straight line for deviation up to 1 MHz.

*3. 5. 3. 8. 2. 8 (U) Overshoot and Droop - 2% maximum, measured at ±300 kHz deviation.

*3. 5. 3. 8. 2. 9 (U) Intermodulation Distortion - Any two tones up to 1 MHz in frequency with a combined deviation of 1 MHz shall not cause intermodulation exceeding -28 db.

*3. 5. 3. 8. 2. 10 (U) Total Harmonic Distortion - 2% maximum for any tone frequency up to 1 MHz at peak deviation of 1 MHz.

*3. 5. 3. 8. 2. 11 (U) Incidental FM - Less than 10 kHz pp.

*3. 5. 3. 8. 2. 12 (U) Incidental AM - Less than 5%.

*3. 5. 3. 8. 2. 13 (U) Modulation Return Grounding - Connected to case internally.

*3. 5. 3. 8. 2. 14 (U) Output Power - 6. 3 watts ±1 db.

*3. 5. 3. 8. 2. 15 (U) Output Impedance - 50 ohms nominal.

*3. 5. 3. 8. 2. 16 (U) Load VSWR - 2. 0:1 maximum (any angle).

*3. 5. 3. 8. 2. 17 (U) Operating Frequency - Any twenty of the ninety assigned IRIG channels from 2200. 5 MHz to 2289. 5 MHz in 1 MHz steps may be utilized; however, this unit shall be programmed for the frequencies listed in paragraph 3. 5. 3. 7. 1.

*3. 5. 3. 8. 2. 18 (U) Frequency Stability - ±0. 003% of assigned frequency; ±0. 005% over environments.

*3. 5. 3. 8. 2. 19 (U) Direct 70 MHz Input - Input level per MIL-STD-188C (+1 dbm into 75-ohm impedance).

*3. 5. 3. 8. 2. 20 (U) Remote Control Channel Selection - The transmitter shall be prewired at factory to tune to the desired channels with a binary, 5-wire code input as specified in table I.

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Table I. Programming Code

Channel Number	RF Freq	Binary Word	Synth Freq	Input Code*				
				A4	A3	A2	A1	A0
41	2212.5 (Red)	1	107.125	0	0	0	0	1
42	2216.5	2	107.325	0	0	0	1	0
43	2220.5 (Green)	3	107.525	0	0	0	1	1
44	2224.5	4	107.725	0	0	1	0	0
45	2228.5	5	107.925	0	0	1	0	1
46	2232.5	6	108.125	0	0	1	1	0
47	2236.5	7	108.325	0	0	1	1	1
48	2240.5 (Yellow)	8	108.525	0	1	0	0	0
49	2244.5	9	108.725	0	1	0	0	1
50	2248.5	10	108.925	0	1	0	1	0
51	2252.5	11	109.125	0	1	0	1	1
52	2256.5 (White)	12	109.325	0	1	1	0	0
53	2260.5	13	109.525	0	1	1	0	1
54	2264.5	14	109.725	0	1	1	1	0
55	2268.5	15	109.925	0	1	1	1	1
56	2272.5	16	110.125	1	0	0	0	0
57	2276.5 (Black)	17	110.325	1	0	0	0	1
58	2280.5	18	110.525	1	0	0	1	0
59	2284.5	19	110.725	1	0	0	1	1
60	2288.5	20	110.925	1	0	1	0	0

All other codes invalid.

*One: 3.5 V ±1.0 V DC (I = 200 ma, max)
Zero: 0 V ±0.4 V DC (I = 6 ma, max)

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*3.5.3.8.2.21 (U) Form Factor - The dimensions of the transmitter shall be in accordance with figure 18.

*3.5.3.8.3 (U) RF Modulator - The RF modulator is a solid-state unit designed for use with the frequency synthesizer to generate 20 factory preset IRIG S-band channels. The unit is also capable of being driven from any vhf source in the 106.525 to 110.975 MHz range that has an output of 20 mw ±1 db with 50 ohm output impedance.

A direct 70-MHz input interface is provided for future growth. This interface permits direct application of an angle modulated signal which is transferred to the S-band output. The interface is compatible with MIL-STD-188C levels and has an input impedance of 75 ohms. Input levels will be +1 dbm.

*3.5.3.8.3.1 (U) Input Power - +28 ±4 volts dc at 2.2 amperes maximum. Internal polarity protection provided.

*3.5.3.8.3.2 (U) Modulation - True FM.

*3.5.3.8.3.3 (C) Output Power - 6.3 watts ± 1 db.

*3.5.3.8.3.4 (U) Output Impedance - 50 ohms nominal.

*3.5.3.8.3.5 (U) Load VSWR - 1.2:1 (any angle).

*3.5.3.8.3.6 (U) Operating Frequency - In accordance with paragraph 3.5.3.8.2.17.

*3.5.3.8.3.7 (U) Frequency Stability - ±0.005% of assigned frequency. ± frequency stability of vhf locking frequency.

*3.5.3.8.3.8 (U) Direct 70-MHz Input - In accordance with paragraph 3.5.3.8.2.19.

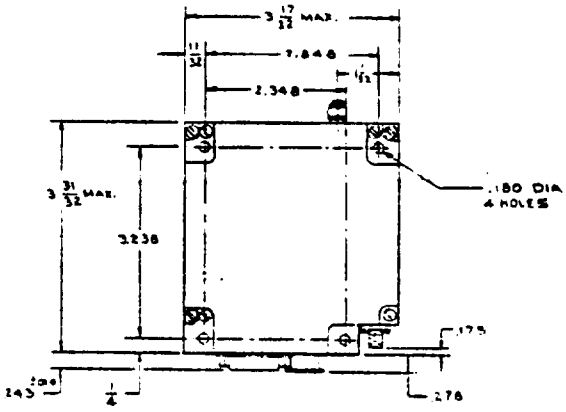
*3.5.3.8.3.9 (U) Form Factor - The dimensions of the rf modulator shall be as shown in figure 19.

*3.5.3.8.4 (U) Frequency Synthesizer - The uhf frequency synthesizer shall be a solid state unit designed for use with the rf modulator to generate 20 factory preset IRIG S-band channels. Channel selection is accomplished by an external control which produces a 5-bit command code.

*3.5.3.8.4.1 (U) Input Power - +28 ±4 volts dc at 600 ±60 milliamperes supplied from the rf modulator.

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CONNECTOR PIN	PIN FUNCTION
A1J1-A	+28 VDC INPUT
A1J1-B	GND
A1J1-C	MOD INPUT
A1J1-D	MOD RET
A1J1-E	70 MHz CONT
A1J1-F	SPARE
A1J1-G	SPARE
A1J2	RF OUTPUT
A1J4	70 MHz INPUT
A2J1-1	2^4 Binary Program Code Input
A2J1-2	2^3
A2J1-3	2^2
A2J1-4	2^1
A2J1-5	2^0
A2J1-6	GND
A2J1-7	SPARE
A2J1-8	SPARE
A2J1-9	SPARE



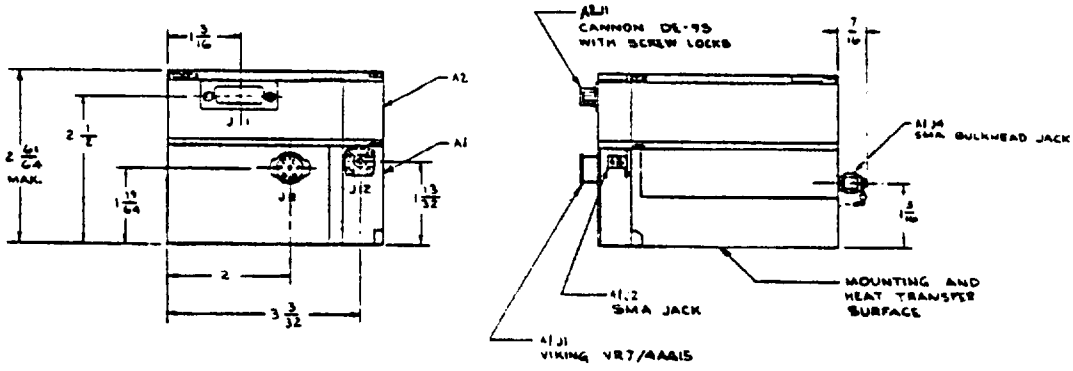
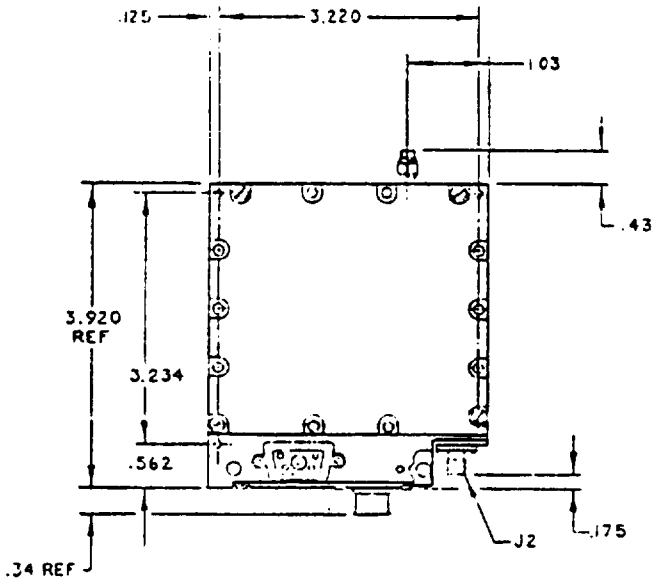
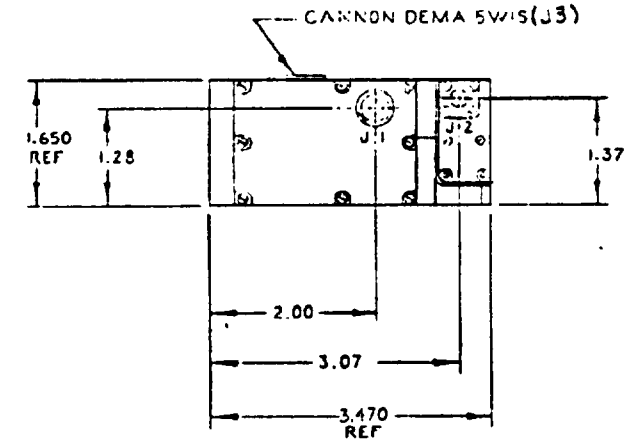


Figure 18. 20-Channel Transmitter, Form Factor and Connector Requirements

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CONNECTOR PIN	PIN FUNCTION
J1-A	+28 VDC INPUT
J1-B	GND
J1-C	MOD INPUT
J1-D	MOD RET
J1-E	70 MHz CONT
J1-F	SPARE
J1-G	SPARE
J2	RF OUTPUT
J3-1	SPARE
J3-2	+28 VDC OUTPUT
J3-3	SPARE
J3-4	28V RET
J3-A1	REF FREQ INPUT
J4	70 MHz INPUT

Figure 19. Transmitter RF Modulator, Form Factor and Connector Requirements

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*3.5.3.8.4.2 (C) Output Frequencies - 20 preselected channels in 25 kHz increments within the 106.525 to 110.975 MHz range to provide the transmitter output rf carrier frequencies as specified in table I.

*3.5.3.8.4.3 (U) Output Power - 15 milliwatts minimum into 50 ohms.

*3.5.3.8.4.4 (U) Spurious Signal Level - Harmonics -15 db; non-harmonics -80 db.

*3.5.3.8.4.5 (U) Frequency Stability - ±0.002% of center frequency.

*3.5.3.8.4.6 (U) Form Factor - The dimensions of the frequency synthesizer shall be as shown in figure 20.

*3.5.3.9 (U) Interstage Isolator - The interstage isolator shall meet the following requirements:

*3.5.3.9.1 (U) Function - The interstage isolator shall be a non-repairable unit that consists of a terminated coaxial circulator, permitting signals to pass in the forward direction, but absorbing them in the reverse direction.

*3.5.3.9.2 (U) RF Power Handling Capabilities - 10 watts minimum forward and reverse.

*3.5.3.9.3 (U) Reverse Isolation - 20 db minimum over 0°C to +70°C; 17 db minimum over -20°C to 0°C and +70°C to +90°C.

*3.5.3.9.4 (U) Frequency Range (Passband) - 2200 to 2300 MHz.

*3.5.3.9.5 (U) Input and Output Impedance - 50 ohms.

*3.5.3.9.6 (U) Voltage Standing Wave Ratio - 1.25:1 maximum over 0°C to 70°C; 1.35:1 maximum over -20°C to 0°C and +70°C to +90°C.

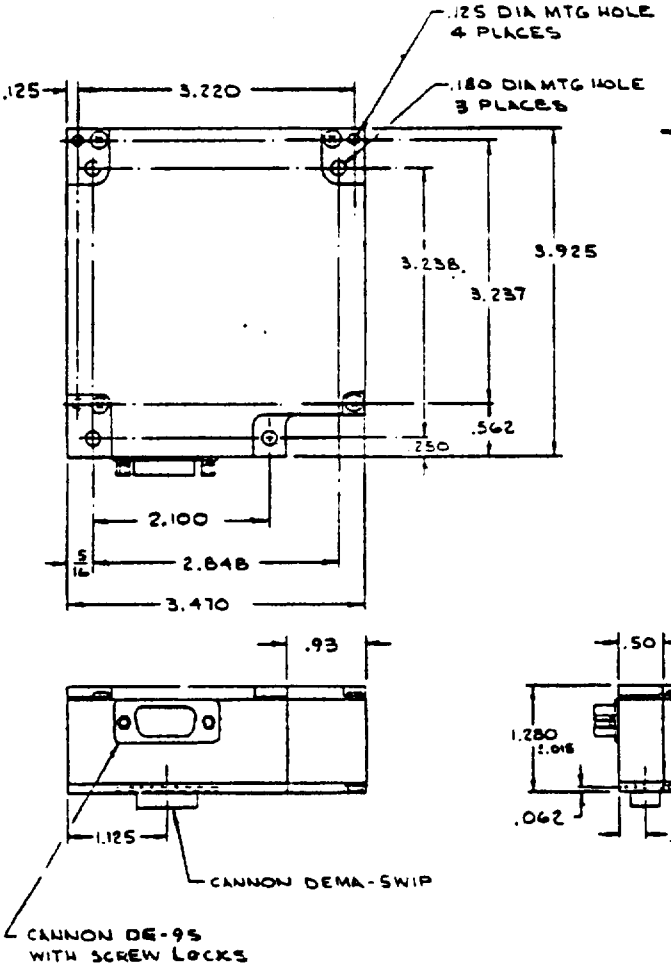
*3.5.3.9.7 (U) Forward Insertion Loss - 0.3db maximum.

*3.5.3.9.8 (U) Form Factor - The dimensions of the isolator shall be as shown in figure 21.

*3.5.3.10 (U) RF Power Amplifier - The rf power amplifier shall meet the following requirements.

*3.5.3.10.1 (U) Functions - The rf power amplifier shall be a factory-repairable unit that provides additional rf amplification of the transmitter output signal.

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CONNECTOR PIN	PIN FUNCTION
J1-1	2^4 Binary Program Code Input
J1-2	2^3
J1-3	2^2
J1-4	2^1
J1-5	2^0
J1-6	GND
J1-7	SPARE
J1-8	SPARE
J1-9	SPARE
J2-1	SPARE
J2-2	+28 VDC INPUT
J2-3	SPARE
J2-4	28V RET
J2-A1	REF FREQ OUT

Figure 20. Transmitter Frequency Synthesizer, Form Factor and Connector Requirements

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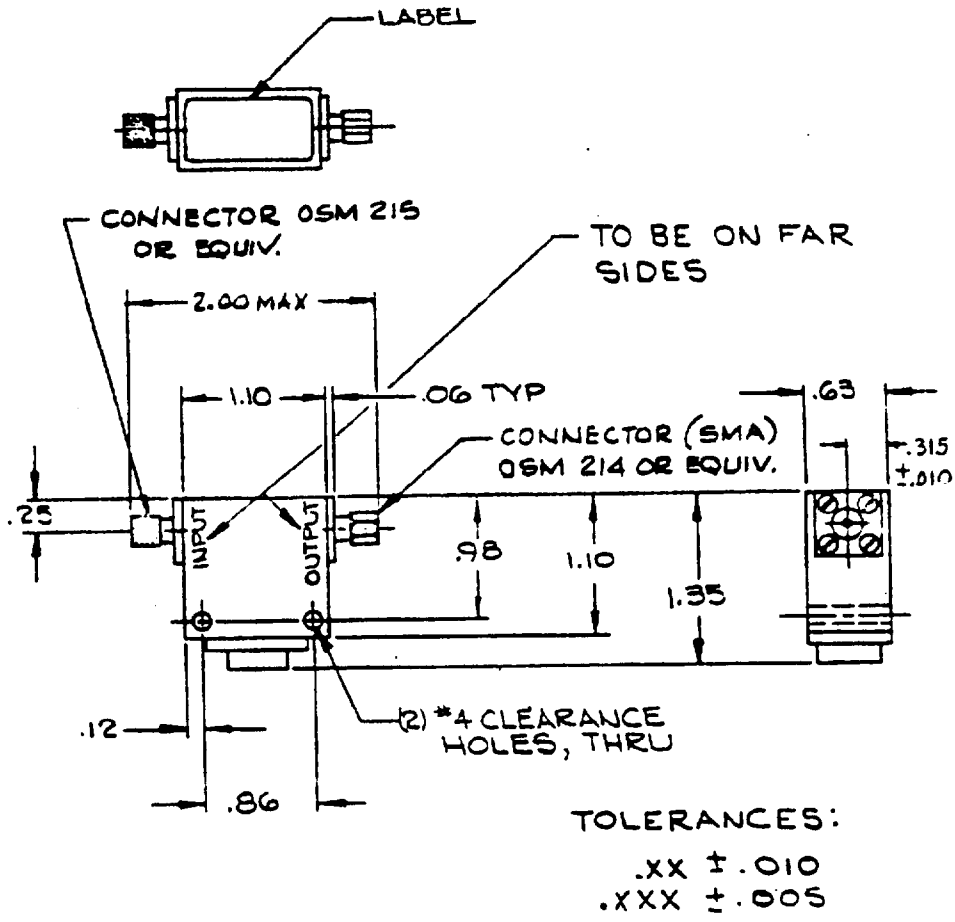


Figure 21. Interstage Isolator, Form Factor and Connector Requirements

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*3.5.3.10.2 (U) Operating Voltage - +24 ±1.0 volts dc at 4.5 amperes maximum.

*3.5.3.10.3 (U) Input/Output Impedance - 50 ohms nominal.

*3.5.3.10.4 (U) Input/Output Frequency Range - 2200 to 2290 MHz.

*3.5.3.10.5 (U) Input VSWR - 1.7:1 maximum.

*3.5.3.10.6 (C) Input RF Power - 6.0 watts, nominal.

*3.5.3.10.7 (C) Output RF Power - 22 watts minimum at output of low pass filter, at an operating baseplate temperature of -20°C to +75°C.

*3.5.3.10.8 (U) Output VSWR - 1.5:1 at any phase angle.

*3.5.3.10.9 (U) Phase Shift and Delay - No discontinuities over the passband.

*3.5.3.10.10 (U) Form Factor - The dimensions of the power amplifier shall be as shown in figure 22.

*3.5.3.10.11 Filter, Low Pass S-Band - The power amplifier shall contain an integral output low pass filter with a 2nd and 3rd harmonic rejection of 40 db minimum; an insertion loss of 0.3 db maximum from 2200 to 2300 MHz; input and output impedance of 50 ohms nominal, and capable of operating with a VSWR of 1.25:1 maximum.

*3.5.3.11 (U) Isolator/Power Monitor - The isolator/power monitor shall meet the requirements specified in paragraph 3.5.1.14 for the T-1220/AKT-22(V) with the following exceptions.

*3.5.3.11.1 (U) RF Power Handling Capability - 40 watts minimum, forward and reflected power.

*3.5.3.11.2 (U) Power Monitor Outputs - 25 ±2.5 microamperes at 20 watts forward or reverse into a 50-microampere meter having an impedance of 20 kilohms. Monitor output terminals shall be isolated from case.

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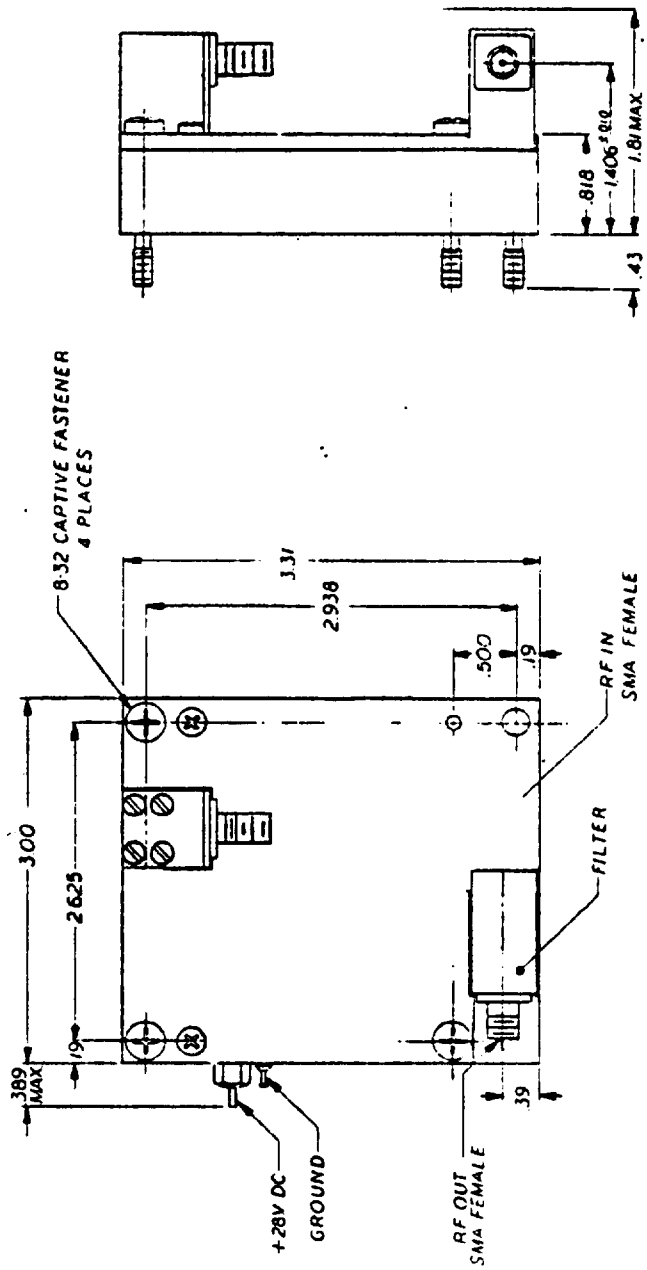


Figure 22. Power Amplifier, Form Factor and Connector Requirements

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*3.5.4 (U) Transmitter-Multiplexer T-1220B/AKT-22(V) - The transmitter-multiplexer shall meet the following requirements.

*3.5.4.1 (U) Function - The T-1220B/AKT-22(V) transmitter-multiplexer shall condition and combine one voice channel and up to eight sonic channels; one voice channel, up to six sonic channels, and up to four trigger signals; or two DIFAR channels into a frequency-division multiplex signal. It shall fm modulate a uhf carrier with the combined multiplex signal and amplify the carrier. The T-1220B/AKT-22(V) shall perform the functions of the T-1220A/AKT-22(V) with increased transmitter power and channel capabilities. The T-1220B/AKT-22(V) shall be convertible to completely or partially perform the functions of the T-1220/AKT-22(V) or T-1295/AKT-22(V) by the deletion, transfer, or replacement of modules. Provision for modulation by an external video or PCM signal shall be provided. The T-1220B/AKT-22(V) shall be IMA repairable.

*3.5.4.2 (U) Form Factor - The transmitter-multiplexer shall be enclosed in a case with dimensions as defined in figure 17.

*3.5.4.3 (U) Weight - The weight of the T-1220B/AKT-22(V) shall not exceed 19 pounds.

*3.5.4.4 (U) Contents - The transmitter-multiplexer shall contain the following major assemblies:

- Seven acoustic voltage controlled oscillators same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.8.
- One wideband acoustic voltage controlled oscillator same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.8.
- One DIFAR voltage controlled oscillator same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.9.
- Seven acoustic low-pass filters same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.10.
- One wideband low-pass filter same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.10.
- One voice-channel low-pass filter same as for the T-1220/AKT-22(V) per paragraph 3.5.1.10.
- One mixer-amplifier same as for the T-1220/AKT-22(V) per paragraph 3.5.1.11.

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- One DIFAR multiplex filter same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.11.
- One 20-channel fm transmitter same as for the T-1295/AKT-22(V) per paragraph 3.5.3.8.
- One interstage isolator same as for the T-1295/AKT-22(V) per paragraph 3.5.3.9.
- One rf power amplifier same as for the T-1295/AKT-22(V) per paragraph 3.5.3.10.
- One isolator/power monitor same as for the T-1295/AKT-22(V) per paragraph 3.5.3.11.
- One DIFAR amplifier adapter per paragraph 3.5.4.8.
- One telemetry mount same as for the T-1220A/AKT-22(V) per paragraph 3.5.2.14.
- One power supply with control circuitry.

*3.5.4.5 (U) Controls - The T-1220B/AKT-22(V shall have the following controls on the front panel.

Circuit Breaker Switch	Applies ac power to unit. Protects unit in the event of overload.
Channel Selector Switch	Provides 5-digit code (ground or open circuit)to transmitter to control transmit frequency for channels 41 through 60. The RMT position allows control of channel from remote switch.
DIFAR Input Switch (behind front panel - visible through window; to be set on installation of unit in aircraft)	In ARR-75 position applies +28 volt dc to DIFAR amplifier adapter to reduce gain for DIFAR input from ARR-75. In ARR-52 position applies ground to DIFAR amplifier adapter to increase gain for output from ARR-52A through AM-6695/AKT-22(V).

*3.5.4.6 (U) Electrical Connections - Connections to external circuits shall be as provided for the T-1295/AKT-22(V) as specified in paragraph 3.5.3.6.

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*3.5.4.7 (U) Overall Unit - The overall transmitter-multiplexer shall meet the requirements for the T-1220A/AKT-22(V) as specified in paragraph 3.5.2.7 with the following exceptions:

*3.5.4.7.1 (U) Operating Frequency - The operating frequency shall be as specified for the T-1295/AKT-22(V) in paragraph 3.5.3.7.1(C).

*3.5.4.7.2 (U) RF Power Output - The rf power output for the T-1220B/AKT-22(V) shall be as specified for the T-1295/AKT-22(V) in paragraph 3.5.3.7.2(C).

*3.5.4.7.3 (U) Signal Inputs - Signal inputs shall be as specified for the T-1220A/AKT-22(V) in paragraph 3.5.2.7.5 with the exception that in the DIFAR mode the T-1220B/AKT-22(V) shall accept either high-level DIFAR signals from the ARR-75 at levels up to 45 volts peak to peak or from the ARR-52 through the AM-6695/AKT-22(V) at levels up to 15 volts peak to peak as specified for the T-1220A/AKT-22(V).

3.5.4.8 (U) DIFAR Amplifier-Module - The DIFAR amplifier-module shall meet the following requirements:

3.5.4.8.1 (U) Function - The DIFAR amplifier-module shall be an IMA repairable unit with dual, isolated input, non-inverting amplifiers to accept DIFAR signals from either dual AN/ARR-75 receivers or AN/ARR-52A receivers utilizing the AM-6695/AKT-22(V) described herein. Mode shall be controlled by an internally mounted switch whose setting shall be visible through the front panel of the transmitter-multiplexer.

• 3.5.4.8.2 (U) Input Power - +28 ±0.5 vdc from the transmitter-multiplexer at 15 milliamperes maximum.

3.5.4.8.3 (U) Input Characteristics - Differential (ground isolated) with 200 kilohms minimum input impedance.

3.5.4.8.4 (U) Input Signal - The amplifiers shall accept the following signals:

3.5.4.8.4.1 (U) ARR-75 Mode - 45 volts peak to peak maximum, differential.

3.5.4.8.4.2 (U) ARR-52 Mode - 15 volts peak to peak maximum (from ARR-52A through AM-6695/AKT-22(V), single-ended.

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3.5.4.8.5 (U) Gain - Unity in ARR-52 mode; 0.325 in ARR-75 mode.

3.5.4.8.6 (U) Output - 15 volts peak to peak, maximum, with specified input.

3.5.4.8.7 (U) Frequency Pass-Band - 10 Hz to 17.4 kHz.

3.5.4.8.8 (U) Form Factor - The dimensions of the DIFAR amplifier-adapter shall be as specified in figure 23.

*3.5.5 (U) Antenna Coupler CU-1497/ARR - The antenna coupler shall meet the following requirements:

3.5.3.1 (U) Function - The antenna coupler shall be an IMA repairable, self-contained multicoupler unit that amplifies and couples vhf-fm signals, modulated with unprocessed data, from the vhf receiving element in the boom antenna to Radio Receiving Set AN/ARR-52A. Each coupler shall contain two identical amplifiers with one input and two outputs each.

*3.5.5.2 (U) Form Factor - The dimensions of the antenna coupler shall be in accordance with figure 24.

3.5.5.3 (U) Weight - The weight of the antenna coupler shall not exceed 3.0 pounds.

*3.5.5.4 (U) Input Power - 28 ±0.5 volts dc, 115 milliamperes, maximum.

*3.5.5.5 (U) Control Voltage - +28 ± 1 volt dc, 45 milliamperes, maximum.

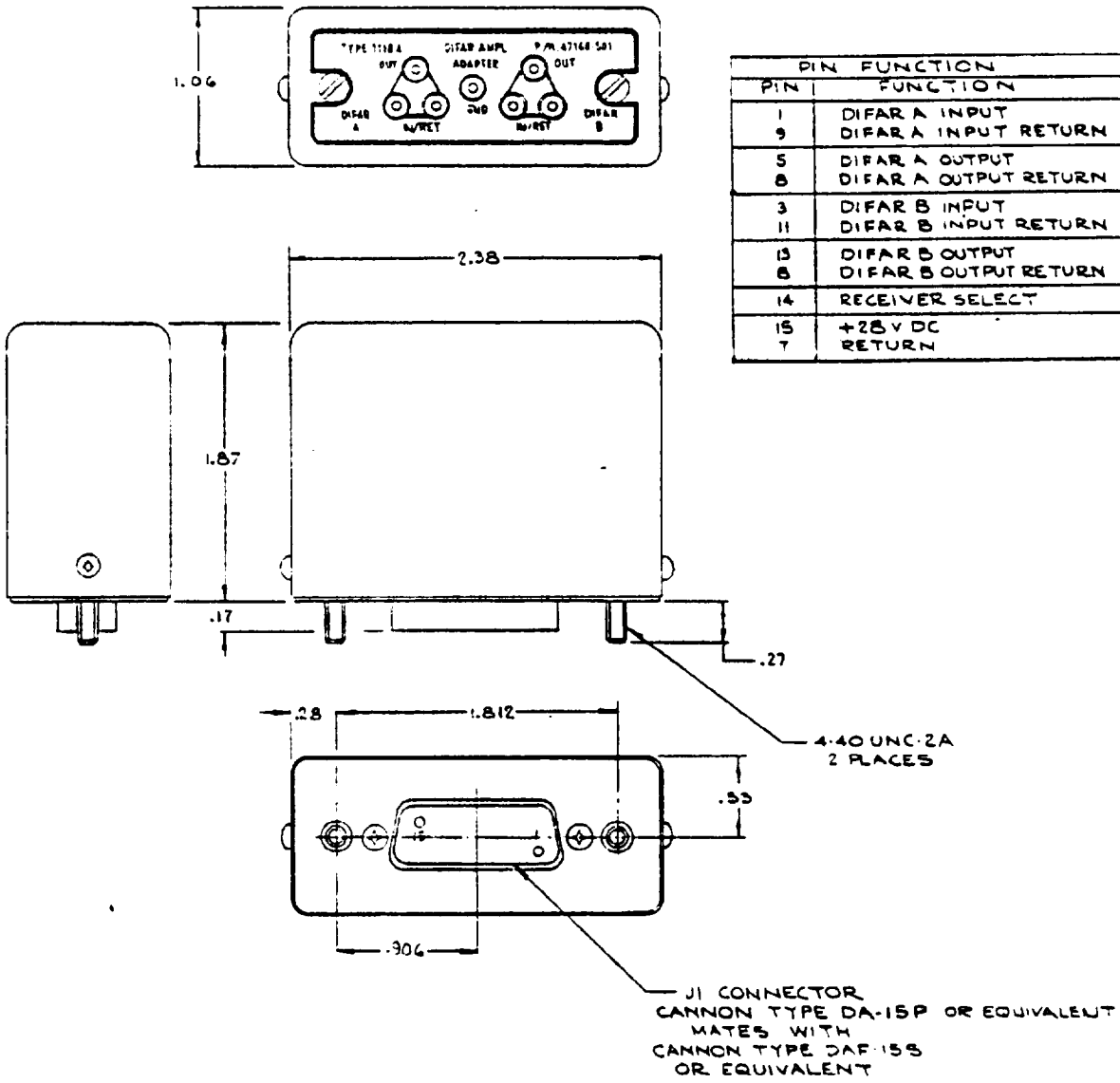
*3.5.5.6 (U) Dynamic Range - A -30 dbm signal at either input shall produce a compression no greater than 1 db at any output.

*3.5.5.7 (U) Frequency Range - 162 to 174 MHz, ±1 db of level at 168 MHz reference. Response down 60 db, minimum from reference at 132 and 204 MHz.

*3.5.5.8 (U) Input/Output Impedance - 50 ohms, nominal.

*3.5.5.9 (U) VSWR - 2.1:1, maximum for inputs (with corresponding outputs terminated in 50 ohms); 1.5:1, maximum for any output (with corresponding input and the other output terminated in 50 ohms).

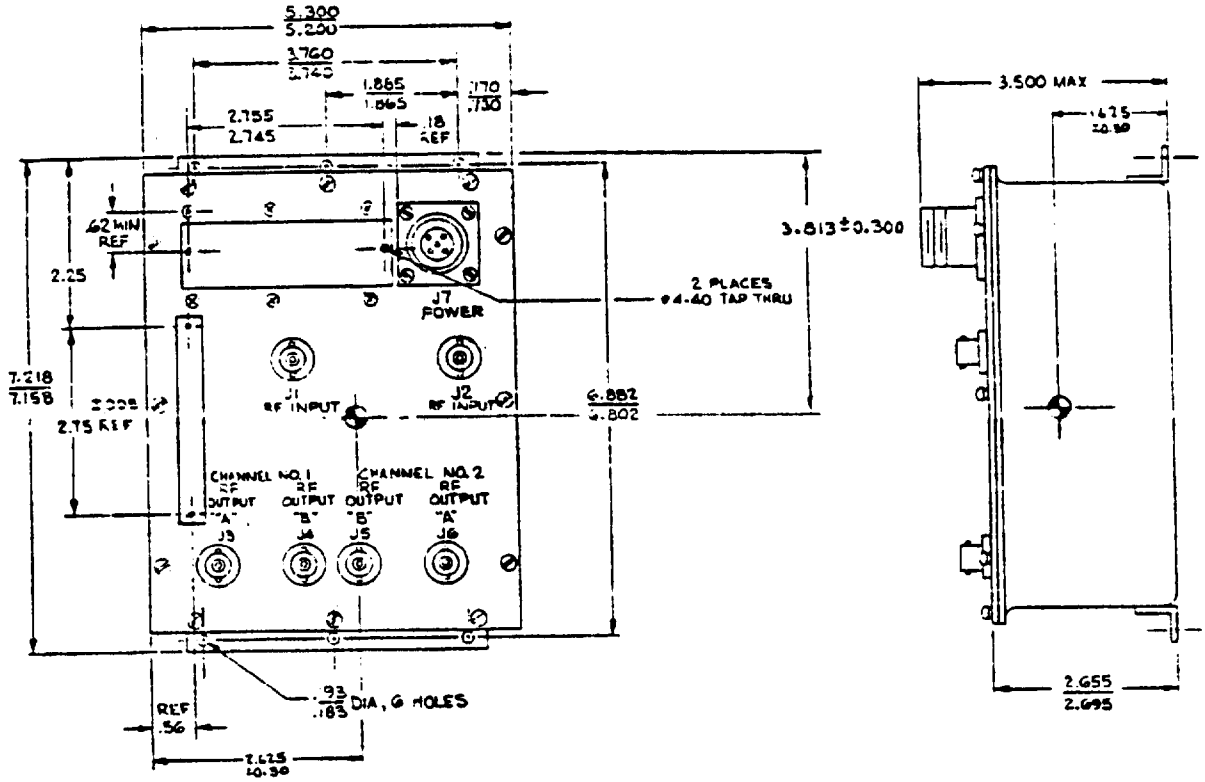
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PIN	FUNCTION
1 9	DIFAR A INPUT DIFAR A INPUT RETURN
5 8	DIFAR A OUTPUT DIFAR A OUTPUT RETURN
3 11	DIFAR B INPUT DIFAR B INPUT RETURN
13 8	DIFAR B OUTPUT DIFAR B OUTPUT RETURN
14	RECEIVER SELECT
15 7	+28 V DC RETURN

Figure 23. DIFAR Amplifier-Adapter, Form Factor and Connector Requirements

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WEIGHT 3 POUNDS, MAXIMUM

ELECTRICAL CONNECTIONS:

REFERENCE DESIGNATION	RECEPTACLE TYPE	FUNCTION
J1 & J2	TYPE C RF CONNECTORS UG-709/U	R.F. INPUTS
J3, J4, J5, & J6	TYPE C RF CONNECTORS UG-709/U	R.F. OUTPUTS
J7	MS3102R-14S-5P OR MS3102E-14S-5P	PIN FUNCTION A: +28VDC LINE B: +28VDC CONTROL C: GND & PWR RETURN D: N.C. E: N.C.

Figure 24. Antenna Coupler CU-1497/ARR, Form Factor and Connector Requirements

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*3.5.5.10 (U) Gain - 8 ± 1 db at 168 MHz reference.

*3.5.5.11 (U) Noise Figure - 5.5 db maximum, over a 0.5 MHz bandwidth within the signal passband.

*3.5.5.12 (U) Electrical Connections - The antenna coupler shall have the following connectors:

Reference Designation	Receptacle Type	Function
J1 and J2	UG-706/U (Type C) connector	RF inputs 1 and 2
J3 and J4	UG-706/U (Type C) connector	RF outputs 1A and 1B
J5 and J6	UG-706/U connector	RF outputs 2A and 2B
J7	MS3102R-14S-5P or MS3102E-14S-5P	Power and control input

*3.5.6 (U) Antenna Coupler CU-2069/ARR - The antenna coupler shall meet the following requirements:

*3.5.6.1 (U) Function - The antenna coupler shall be an IMA repairable self-contained multicoupler unit that amplifies and couples vhf-fm signals, modulated with unprocessed data from the vhf receiving element in the antenna to one or two Radio Receiving Sets AN/ARR-52A. The coupler shall have one input and four isolated outputs.

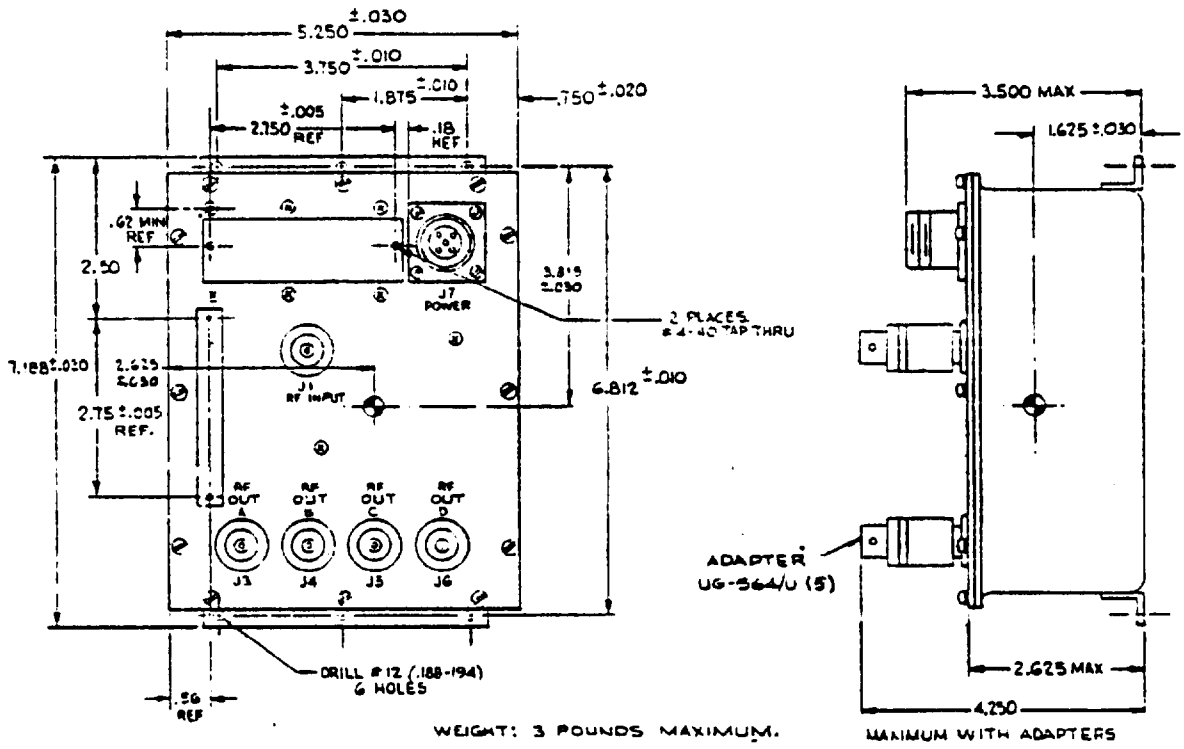
*3.5.6.2 (U) Form Factor - The dimensions of the antenna coupler shall be in accordance with figure 25.

*3.5.6.3 (U) Weight - The weight of the antenna coupler shall not exceed 3.3 pounds.

*3.5.6.4 (U) Input Power - 28 ± 1.0 volt dc, 100 milliamperes, maximum.

*3.5.6.5 (U) Control Voltage - 28 ± 1 volt dc, 50 milliamperes, maximum.

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ELECTRICAL CONNECTIONS:

REFERENCE DESIGNATION	RECEPTACLE TYPE	FUNCTION	MATES WITH
J1	GREF 1C28-0001 N TYPE FEMALE	RF INPUT	UG-710( )/U USING ADAPTER UG-564/U
J3,J4,J5,J6	GREF 1028-0001 N TYPE FEMALE	RF OUTPUTS A,B,C,&D	
J7	MS3102R-14S-5P OR MS3102E-14S-5P	POWER INPUT	GC15A-4A14S-5S

PIN CONNECTIONS OF J7

A- +28 VDC LINE
B- +28 VDC CONTROL
C- GRD & PWR RETURN
D- NO CONNECTION
E- NO CONNECTION

Figure 25. Antenna Coupler CU-2069/ARR, Form Factor and Connector Requirements

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*3.5.6.6 (U) Dynamic Range - A -27 dbm signal at the input shall cause no greater than 1 db compression at output.

*3.5.6.7 (U) Bandwidth - 162 to 174 MHz ±1 db of level at 168 MHz reference. Response down 60 db, minimum at 128 and 208 MHz.

*3.5.6.8 (U) Input/Output Impedance - 50 ohms, nominal.

*3.5.6.9 (U) VSWR - 1.5:1, maximum for input (with all outputs terminated in 50 ohms); 1.5:1 maximum for outputs (with input and all other outputs terminated in 50 ohms).

*3.5.6.10 (U) Gain - 8 ± 1 db at 168 MHz reference.

*3.5.6.11 (U) Rejection - Input signals at level of +19 dbm in range from 225 to 400 MHz shall be down 100 db at each output.

*3.5.6.12 (U) Intermodulation Distortion - -80 dbm, maximum at the output for any two input signals with equal amplitude and combined power level of -27 dbm in the range from 162 to 174 MHz.

*3.5.6.13 (U) Noise Figure - 4.3 db, maximum over a 0.5 MHz bandwidth within the signal passband.

*3.5.6.14 (U) Isolation - 40 db, minimum between any of the four inputs.

*3.5.6.15 (U) Electrical Connections - The antenna coupler shall have the following connectors:

Reference Designation	Receptacle Type	Function
J1	UG-706/U connector (Type C)	RF input
J3 through J6	UG-706/U connector (Type C)	RF outputs
J7	MS3102E-14S-5P or MS3102R-14S-5P	Power and control input

3.5.7 (U) Antenna AS-2744/AKT-22(V) - The antenna shall meet the following requirements:

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3.5.7.1 (U) Function - The antenna shall be a fuselage-mounted, retractable boom antenna assembly used in the telemetric transmitting set to receive rf fm signals in the 162 to 174 MHz band and transmit rf signals in the 2200 to 2290 MHz band (S-band).

*3.5.7.2 (U) Form Factor - The dimensions of the antenna boom shall be in accordance with figure 26. The antenna and actuator shall weigh less than 21 pounds.

*3.5.7.3 (U) Contents - The antenna shall contain a vhf receiving element, a uhf transmitting element, and an electrically operated retractor assembly as described in paragraph 3.5.9 used to extend and retract the antenna boom through an angle of approximately 90 degrees. The retractor assembly shall contain a retractor mechanism, a reversible motor with a magnetically operated brake, two limit switches and positions indicating circuitry.

*3.5.7.4 (U) Input Power - 28 +1, -4 vdc, 3.0 amperes maximum antenna actuate power).

3.5.7.5 (U) Full Actuation Time - 30 seconds.

*3.5.7.6 (U) Transmitting Frequency Band - 2200 to 2290 MHz ±1 db.

*3.5.7.7 (U) Receiving Frequency Band - 162 to 174 MHz ± 1 db.

3.5.7.8 (C) Polarization - Vertical polarization in both transmit and receive modes.

3.5.7.9 (U) VSWR - 1.50:1 for transmit range, 2,00:1 for receive range.

3.5.7.10 (C) Gain - 7 db minimum for transmit band, 0 db for receive band,

*3.5.7.11 (C) Transmit Elevation Beam Width - The transmit antenna element shall have a minimum 3 db elevation beamwidth of ten degrees centered at the horizon.

*3.5.7.12 (C) Transmit Azimuth Radiation Pattern - The transmit antenna element shall have an omnidirectional radiation pattern in the azimuth plane.

*3.5.7.13 (C) Receive Antenna Pattern - The receive antenna element shall have an omnidirectional pattern.

*3.5.7.14 (C) RF Power Handling Capacity - Up to 100 watts, transmit element.

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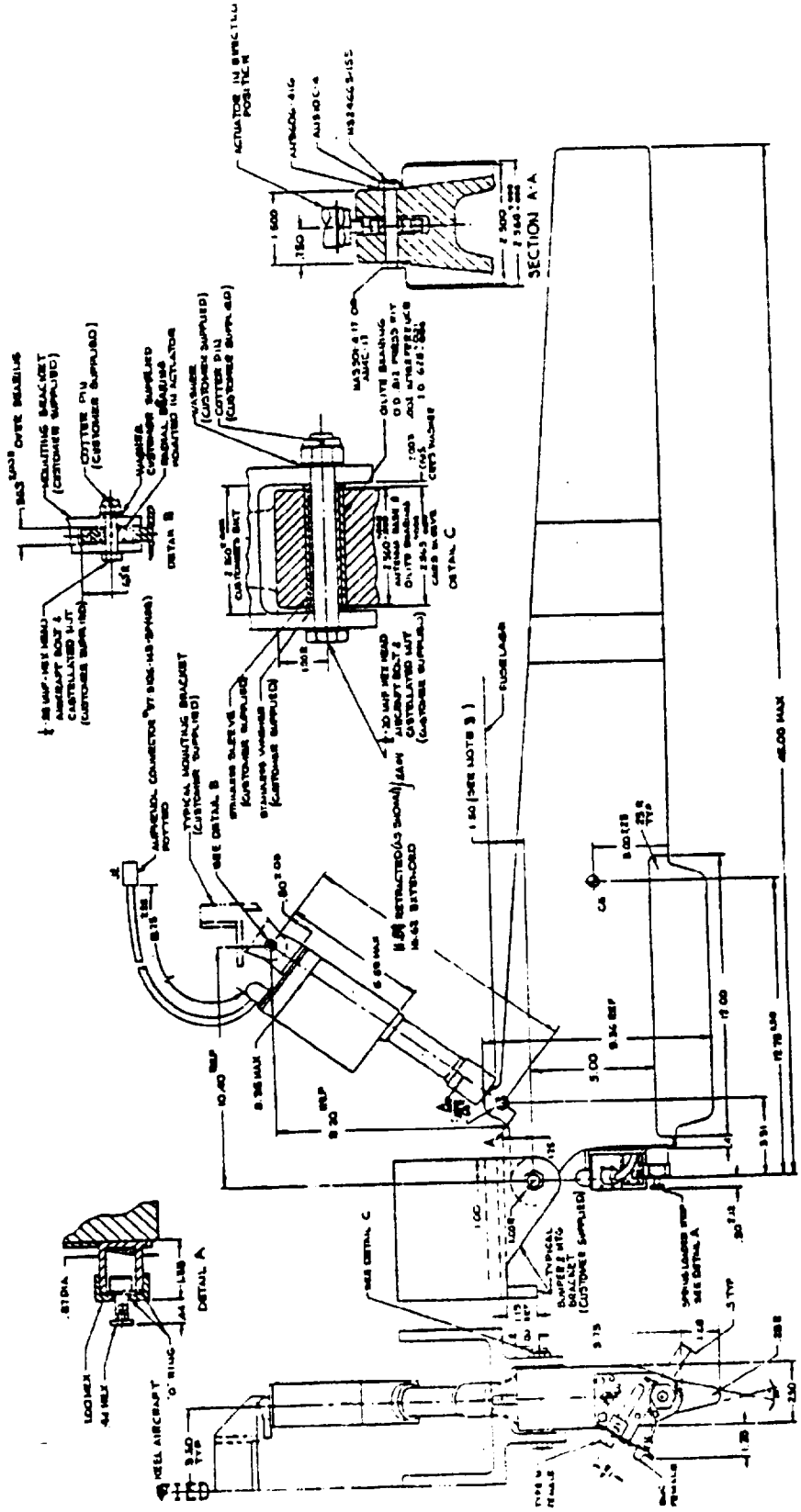


Figure 24. Antenna AS-2744/AKT-22, Form Factor

Notes
(1) For new production order, bevel inside edges of section A-A clevis to depth of 0.25 inches and width of 0.25 inches. length of bevel equal to arc between 60° and 150° with origin at bolt hole center using longitudal axis of antennas as reference. Surface beyond this arc to be machined for smooth merge to existing contour.

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*3.5.7.15 (U) Electrical Connections - The antenna shall have the following electrical connectors:

Reference Designations	Receptacle Type	Function
J1	N	Input to transmit element
J2	97-3106-14S-5P(43B)	Actuator control and power
J3	BNC	Output from receive element

*3.5.8 (U) Antenna, AS-3033/AKT-22(V). - The antenna shall meet the following requirements:

*3.5.8.1 (C) Function - The antenna shall be a non-repairable, frangible fuselage-mounted, retractable boom assembly used to receive rf energy in the band from 162 to 174 MHz from sonobuoys and radiate rf energy in the band from 2200 to 2290 MHz (S-band).

*3.5.8.2 (C) Transmitting Frequency Band - 2200 to 2290 MHz.

*3.5.8.3 (C) Receiving Frequency Band - 162 to 174 MHz.

*3.5.8.4 (C) Polarization - Vertical polarization in both transmit and receive modes when extended 90 degrees from fuselage.

*3.5.8.5 (U) Voltage Standing Wave Ratio (VSWR) - 1.50:1 for transmit band, 2.00:1 for receive band.

*3.5.8.6 (C) Gain - 5 db minimum for transmit band; 0 db minimum for receive band.

*3.5.8.7 (U) Coupling - Signal coupling from the transmitting antenna to the receiving antenna shall be less than 70 db.

*3.5.8.8 (C) Transmitting Power Capacity - 100 watts maximum.

*3.5.8.9 (C) Transmit Elevation Beam Width - The transmit antenna element shall have a minimum 3 db elevation beam width of 20 degrees centered at the horizon.

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*3.5.8.10 (C) Transmit Azimuth Radiation Pattern - The transmit antenna element shall have an omnidirectional radiation pattern (within ± 1-1/4 db) in the azimuth plane.

*3.5.8.11 (U) Phase Center - The phase center of the transmitting antenna element shall be at a minimum of 36 inches from the aircraft surface when in operate position.

*3.5.8.12 (C) Receive Antenna Pattern - The receive antenna element shall have an omnidirectional pattern in the azimuth plane.

*3.5.8.13 (U) Aerodynamic Loading - The antenna shall be capable of sustaining the following steady state air loads:

Vertical - negligible

Lateral, fore and aft - 64 pounds at the tip, 112 pounds at the center of pressure.

*3.5.8.14 (U) Frangibility - The antenna shall fracture in the extended position, at a point no greater than 14 inches from the center of the pivot bearing when subjected to the following loads:

Vertical - 1500 pounds

Lateral, fore and aft - 256 pounds at the tip.

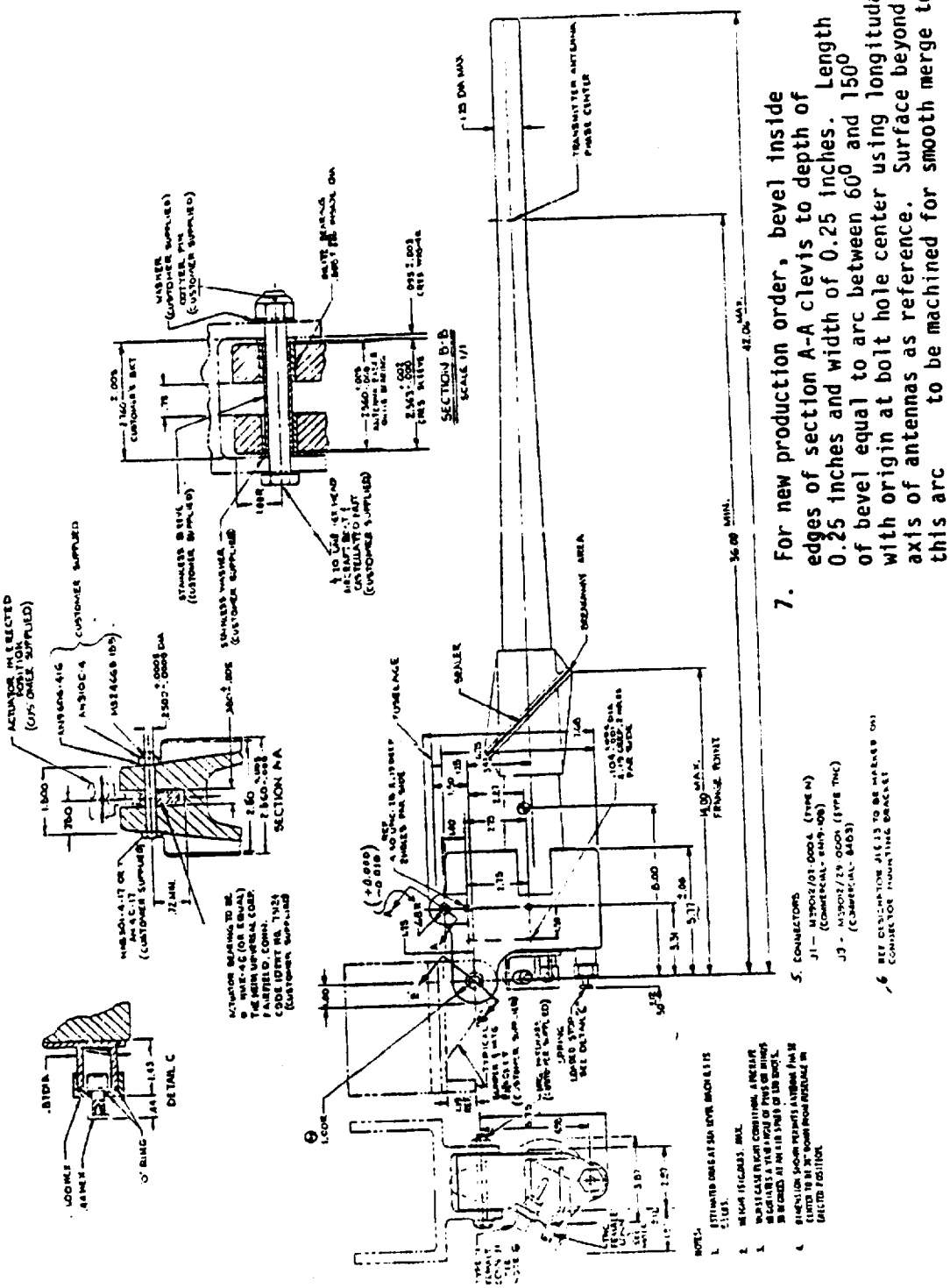
*3.5.8.15 (U) Weight - The antenna shall weigh less than 10 pounds.

*3.5.8.16 (U) Electrical Connections - The antenna shall have the following electrical connectors:

Reference Designation	Receptacle Type	Function
J1	M39012/ 02-0004 Type N	Input to transmit element
J3	M39012/ 29-0001 Type TNC	Output from receive element

*3.5.8.17 (U) Form Factor - The dimensions of the antenna shall be as shown in figure 27.

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7. For new production order, bevel inside edges of section A-A clevis to depth of 0.25 inches and width of 0.25 inches. Length of bevel equal to arc between 60° and 150° with origin at bolt hole center using longitudal axis of antennas as reference. Surface beyond this arc to be machined for smooth merge to existing contour.
SECTION A-A
SECTION B-B
DETAIL C
TRANSMITTER ANTENNA PHASE CENTER
BRACKET AREA
FUSELAGE
FRONT POINT
5. CONNECTORS
J1 - M39012/01-0004 (TYPE N) (COMMERCIAL-8449-006)
J3 - M39012/19-0004 (TYPE TNC) (COMMERCIAL-8463)
6. REF DESIGNATION J1 & J3 TO BE MARKED ON CONNECTOR MOUNTING BRACKET
NOTES:
1. ESTIMATED DRAG AT 350 KNOTS, SEA LEVEL, STD. DAY, CLEAN CONFIGURATION, 0.0 DEGREES ANGLE OF ATTACK, 0.0 DEGREES ANGLE OF SIDESLIP, 0.0 DEGREES ANGLE OF ROLL, 0.0 DEGREES ANGLE OF PITCH, 0.0 DEGREES ANGLE OF YAW

*3.5.9 (U) Actuator, Electromechanical Linear, TG-229/AKT-22(V) - The actuator shall meet the following requirements:

*3.5.9.1 (U) Function - The actuator shall be an IMA repairable electromechanical assembly utilized to remotely erect and retract the antenna. It shall contain a retractor mechanism, a reversible motor with a magnetically operated brake, two limit switches and position indicating circuitry.

*3.5.9.2 (U) Operating Loads - Normal operating load 270 pounds tension or compression; maximum operating load 540 pounds tension or compression.

*3.5.9.3 (U) Operating Stroke - As shown in figure 28.

*3.5.9.4 (U) Operating Speed - 0.20 inches per second ±20% at normal operating load with 26 volts dc applied at 70°F.

*3.5.9.5 (U) Static Load - Ultimate static load 3780 pounds tension or compression. Limit static load 1800 pounds tension or compression.

*3.5.9.6 (U) Operating Life - 10,000 full stroke cycles at rated load and duty cycle.

*3.5.9.7 (U) Duty Cycle - 1 minute ON; 5 minutes OFF at rated load.

*3.5.9.8 (U) Voltage Requirement - Normal rated voltage 26 volts dc. Unit must operate over voltage range 24 to 29 volts dc.

*3.5.9.9 (U) Current Requirement - Normal operating current 2.5 amperes, maximum operating current 4.5 amperes.

*3.5.9.10 (U) Limit Switches - Actuator shall contain internally adjustable limit switches set to limit stroke as shown in figure 28.

*3.5.9.11 (U) Mechanical Stops - Actuator shall contain internal non-jamming mechanical stops to limit overtravel in the event of a switch malfunction.

*3.5.9.12 (U) Overtravel - Actuator overtravel shall be held to a minimum by means of a separate motor brake.

*3.5.9.13 (U) Direction of Operation - Actuator shall be electrically reversible.

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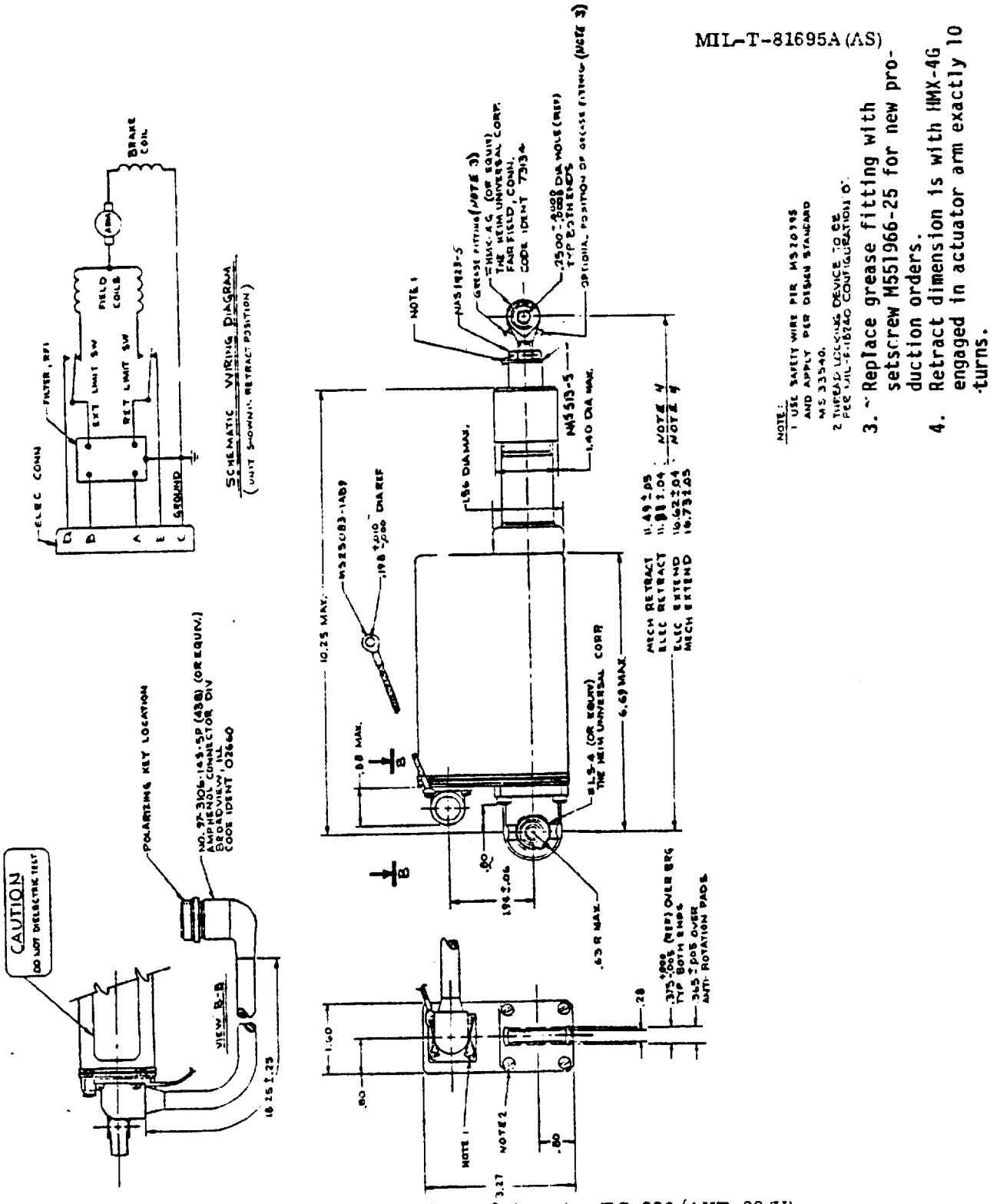


Figure 28. Linear Electro-Mechanical Actuator TG-229/AKT-22(V), Form Factor

NOTE:
1. USE SAFETY WIRE PER MS20995 AND APPLY PER DESIGN STANDARD MS 33540.
2. THREAD LOCKING DEVICE TO BE PER MIL-F-18240 CONFIGURATION D.
3. Replace grease fitting with setscrew M551966-25 for new production orders.
4. Retract dimension is with HMX-4G engaged in actuator arm exactly 10 turns.

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*3.5.9.14 (U) Reversibility - Actuator shall not reverse as a result of application of external loads up to the maximum static load combined with vibration and/or shock.

*3.5.9.15 (U) Endplay - Actuator endplay shall not exceed .010 inch TIR when measured under a 10 pound reversing load.

*3.5.9.16 (U) Anti-Rotation Device - Actuator shall contain an anti-rotation device to prevent rotation of extending ram.

*3.5.9.17 (U) Lubrication - Actuator shall be lubricated for rated life at time of manufacture.

*3.5.9.18 (U) Weight - The actuator shall weigh 4.0 pounds maximum.

*3.5.9.19 (U) Form Factor - The dimension of the actuator shall be as shown in figure 28.

*3.5.10 (U) Control-Indicator C-8988/AKT-22(V) - The control-indicator shall meet the following requirements:

*3.5.10.1 (U) Function - The control-indicator shall be an IMA repairable, self-contained, separately housed unit that serves as the central operating and monitoring facility in the transmitting system. The control-indicator shall perform the following functions: panel lighting control, operating mode control, voice control, antenna actuation control and rf power monitoring.

*3.5.10.2 (U) Form Factor - The dimensions of the control-indicator shall be in accordance with figure 29.

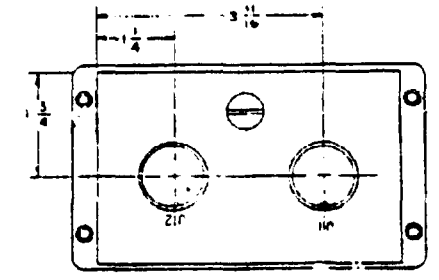
*3.5.10.3 (U) Weight - The weight of the control-indicator shall not exceed 3.0 pounds.

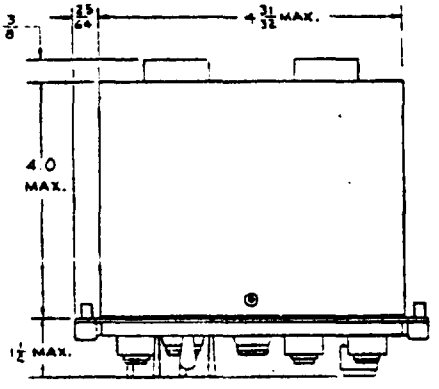
*3.5.10.4 (U) Contents - The control-indicator shall contain the following circuits: panel lighting, operating mode control, sonar trigger channel selection, transmitter power monitoring, voice control, antenna actuation control.

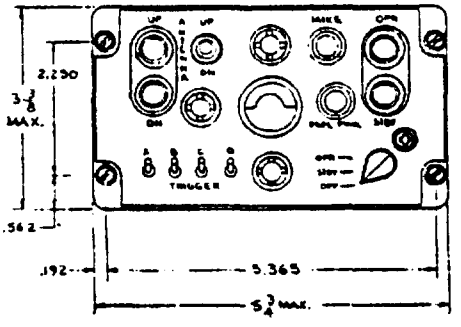
*3.5.10.4.1 (U) Panel Lighting - Panel illumination shall be provided by three panel indicators which shall be edge lit by 28-volts maximum from the main instrument control panel through the radio receiving set. The edge lit panel shall be in accordance with MIL-P-7788.

*3.5.10.4.2 (U) Operating Mode Control - A complete power off mode, standby mode, and operated mode shall be provided. In the standby and operate modes 28 volts dc shall be applied to transmitter-multiplexer to energize the power supply circuits and the multiplexer. In the operate mode 28-volts dc shall be applied to transmitter-multiplexer to operate a relay and energize the transmitter. A fourth position for DIFAR enable shall be provided and wired but unmarked.

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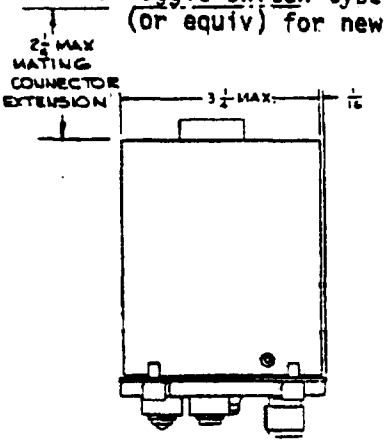
NOTE:

Tolerances unless noted:

Fractions: ±1/64
Decimals: ±0.005

NOTES:

1. CONNECTORS
   J1 TYPE MS3130E-27-50P MATES WITH MS3137P-27-50S
   J2 TYPE MS3130E-37-50P MATES WITH MS3137-37-50S
2. FRONT AND REAR PANEL SYMBOLIZATION ⅛ HIGH.
3. MOUNTING FASTENERS PER MIL-F-25179.
4. FRONT PANEL KNOB IS MS91528-01PB.
5. Toggle switch type is C and K (P/N 7201SYZ) (or equiv) for new production orders.



PIN CONNECTIONS OF

J1 PIN	J1 FUNCTION
1	RCVR A STANDARD OUTPUT
2	RCVR A OUTPUT RETURN
3	RCVR A OUTPUT SHIELD
4	RCVR B STANDARD OUTPUT
5	RCVR B OUTPUT RETURN
6	RCVR B OUTPUT SHIELD
7	RCVR C STANDARD OUTPUT
8	RCVR C OUTPUT RETURN
9	RCVR C OUTPUT SHIELD
10	RCVR D STANDARD OUTPUT
11	RCVR D OUTPUT RETURN
12	RCVR D OUTPUT SHIELD
13	MCJR TRIG RCVR A
14	MCJR TRIG RCVR A RETURN
15	MCJR TRIG RCVR A SHIELD
16	MCJR TRIG RCVR B
17	MCJR TRIG RCVR B RETURN
18	MCJR TRIG RCVR B SHIELD
19	MCJR TRIG RCVR C
20	MCJR TRIG RCVR C RETURN
21	MCJR TRIG RCVR C SHIELD
22	MCJR TRIG RCVR D
23	MCJR TRIG RCVR D RETURN
24	MCJR TRIG RCVR D SHIELD
25	N. C.
26	N. C.
27	N. C.

J2 PIN	J2 FUNCTION
1	POWER RETURN
2	FORWARD POWER
3	REFLECTED POWER
4	TRANS (CONT)
5	28 V DC (STBY)
6	IND (ON)
7	IND (STBY)
8	MCJR VOICE OUTPUT
9	MCJR VOICE RETURN
10	ICS INPUT
11	ICS RET
12	ICS OUTPUT } 1 AIC-14
13	ICS KEY }
14	+28V DC INPUT
15	DC GROUND
16	PANEL LIGHTS
17	UP IND ANT
18	DOWN IND ANT
19	ANTENNA UP
20	ANTENNA DOWN
21	ANTENNA UP(ON SIDE OF WL)
22	N.C.
23	N.C.
24	N.C.
25	N.C.
26	DIMM OFF/ON
27	+28V (UP SIDE WHEEL LOCK
28	N.C.
29	N.C.
30	N.C.
31	N.C.
32	N.C.
33	N.C.
34	N.C.
35	N.C.
36	N.C.
37	N.C.

Figure 29. Control-Indicator C-8988/AKT-22(V) Form Factor and Connector Requirements

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3.5.10.4.3 (U) Sonar Trigger Channel Selection - The sonar trigger channel selection circuit shall control the application of the receiver channel inputs to the multiplexer.

3.5.10.4.4 (U) Transmitter Power Monitoring - The transmitter power monitoring circuits shall allow both reflected and forward rf power output of the fm transmitter to be monitored on a power meter when the transmitter-multiplexer is in the operate mode. A power minimum indication shall be provided at the center of the meter.

3.5.10.4.5 (U) Voice Control - The voice control circuit shall interconnect the aircraft Interphone Communications System to enable the operator to send voice transmissions to the shipboard receiving system, or talk over the aircraft intercom system.

3.5.10.4.6 (U) Antenna Actuation Control - The antenna actuation control circuit shall enable the antenna boom to be extended or retracted and indicate its position.

*3.5.10.5 (U) Controls - The control-indicator shall have the following controls:

ANTENNA UP-DN Switch	Controls actuation of Antenna AS-2744/AKT-22(V) (boom antenna). In UP position, switch routes +28 vdc to corresponding control elements in antenna actuator circuit to retract boom antenna. In position DN, switch routes +28 vdc to corresponding control elements in antenna actuator circuit to extend antenna. (Note that a wheel interlock circuit in the aircraft disables the operation of this switch if the aircraft wheels are down.)
MIKE Pushbutton Switch	When pressed, activates aircraft Interphone Communications System (ICS) key and routes ICS voice inputs to transmitter-multiplexer for transmission to shipboard telemetry receiving system. (ICS voice transmissions are routed only to aircraft intercom system when pushbutton is in normal position.)
REFL PWR Pushbutton Switch	When pressed and held, applies reflected rf power output from isolator AT1 to rf power monitor meter M1. In normally closed position forward rf power is applied directly to M1.

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TRIGGER A, B, C, and D Switches	Used, in "up" position, to individually route sonar trigger inputs from receiver (A, B, C, or D) standard output to transmitter-multiplexer. In "down" position (off) switches effectively short corresponding multiplexer inputs.
OFF-STBY-OPR Switch	A ganged, four-position selector switch that controls the turnoff, turnon, and operation of the transmitter-multiplexer and the telemetric transmitting set. In OFF position, transmitter-multiplexer is deenergized. In STBY position, applies the +28 vdc standby command voltage transmitter-multiplexer power control circuit to energize transmitter-multiplexer in the standby operating mode. In OPR position, maintains the standby command voltage and applies the +28 vdc operate command voltage to transmitter-multiplexer power control circuit to energize the transmitter-multiplexer in the operate mode. The fourth unmarked position shall be wired for DIFAR operation.

3.5.10.6 (U) Electrical Connection - The control-indicator shall have the following electrical connectors:

Reference Designation	Receptacle Type	Function
J1	MS3130E27-50P	Sonar Trigger Inputs and Outputs
J2	MS3130E37-50P	Power and Control Signals

*3.5.11 (U) Control-Indicator C-8988A/AKT-22(V) - The control-indicator shall meet the requirements for Control-Indicator C-8988/AKT-22(V) as specified in paragraph 3.5.10 with the following changes and additions:

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*3.5.11.1 (U) Function - The C-8988A/AKT-22(V) shall perform the same function as the C-6988/AKT-22(V) with additional functions of sonar trigger channel filtering and line-impedance matching and control to disable subcarrier channels in the transmitter-multiplexer to prevent interference with the trigger signals.

*3.5.11.2 (U) Form Factor - The dimensions of the control-indicator shall be as defined in figure 30.

*3.5.11.3 (U) Contents - The control-indicator shall contain, in addition to the circuits specified in paragraph 3.5.10.4 for the C-8988/AKT-22(V), trigger-filter and line driver circuits.

*3.5.11.3.1 (U) Operating Mode Control - A marked DIFAR mode shall be included in addition to the modes specified in paragraph 3.5.6.4.2. In the DIFAR mode the control shall apply the standby control voltage to the transmitter-multiplexer as in STBY and OPR modes, 28 volts to the transmitter-multiplexer to operate a relay and energize the transmitter, and 28 volts to the DIFAR filter in the transmitter-multiplexer to energize a relay which shall select DIFAR channel signals instead of the multiplexer output for application to the transmitter.

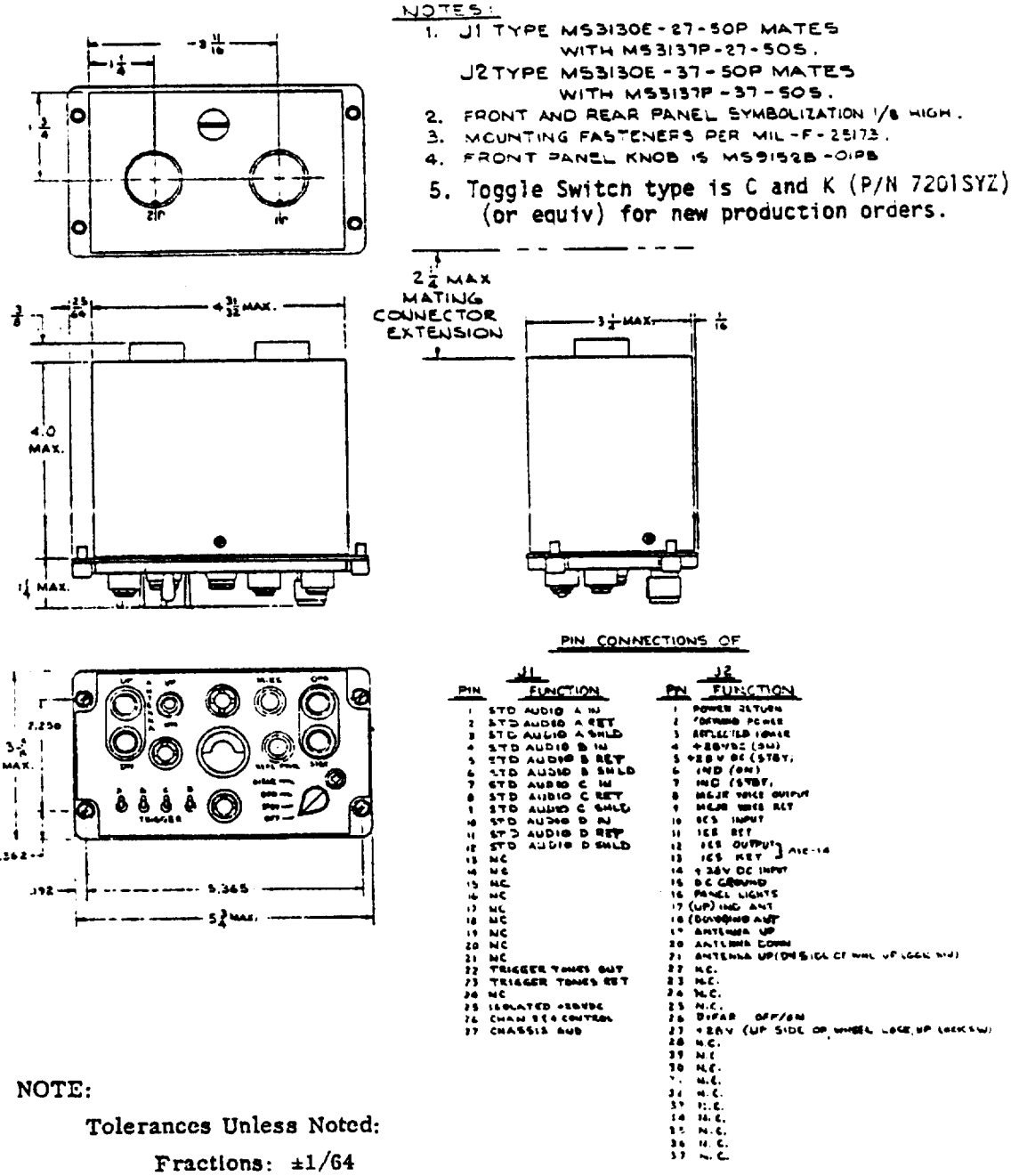
*3.5.11.3.2 (U) Sonar Trigger Channel Selection - The sonar trigger channel selection circuit shall control the application of the receiver channel inputs to the multiplexer and shall disable subcarrier channels in the trigger tone band.

*3.5.11.3.3 (U) Trigger-Filter and Line Driver - The C-8988A/AKT-22(V) shall have a trigger-filter as described in paragraph 3.5.1.12 except for connectors. The output of the trigger-filter shall feed a emitter-follower transistor line-driver circuit to match impedance to the transmission line to the transmitter-multiplexer. The characteristics of the composite trigger signals at the input to T-1220A/AKT-22(V) shall be the same as those at the output of the trigger-filter in the T-1220/AKT-22(V).

*3.5.11.4 (U) Controls - The control-indicator shall have the controls specified in paragraph 3.5.10.5 with the following differences:

TRIGGER A, B, C, and D Switches	When set to on (up) position, these switches individually select standard audio output (trigger tones) from receiver channels and apply them to input of trigger-filter and line driver section of control-indicator for subsequent application to input of transmitter-multiplexer. Also in the on (up) position, the isolated +28 volt dc power input to the low-pass filters for the 29.2 and 37.5 kHz VCO subchannels is interrupted to prevent interference between signals on these channels with the trigger signals. In the off (down) position the trigger tone input is disconnected and power is supplied to operate the 29.2 and 37.5 kHz channels.

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NOTE:

Tolerances Unless Noted:

Fractions: ±1/64
Decimals: ±0.005

Figure 30. Control-Indicator C-8988A/AKT-22(V) Form Factor and Connector Requirements

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Operating Mode Switch	A ganged four-position selector switch that controls the turnoff, turnon, and operation of the transmitter-multiplexer and the telemetric transmitting set. In OFF position, transmitter-multiplexer is deenergized. In STBY position, applies the +28 volt standby command voltage to transmitter-multiplexer power control circuit to energize transmitter-multiplexer in the standby operating mode. No rf output. In OPR position, maintains the standby command voltage and applies the +28 volt operate command voltage to transmitter-multiplexer power control circuit to energize the transmitter-multiplexer in the operate mode. RF output enabled. In DIFAR position, maintains transmitter-multiplexer energized in power operate mode, but with the DIFAR signals being applied to the transmitter.

*3.5.12 (U) Amplifier AM-6695/AKT-22(V) - The amplifier shall meet the following requirements:

*3.5.12.1(U) Function - The amplifier shall be an IMA repairable solid-state unit which shall be plugged into front panel test jacks on selected R-1170/ARR-52 receivers to enable the system to handle DIFAR signals. It shall differentially detect presence of AN/SSQ-53 Sonobuoy (DIFAR) transmissions at discriminator test point of the receivers and shall accept composite DIFAR signals from the receiver channel and convert the signals to low-impedance unbalanced output signals for application to the transmitter-multiplexer. It shall serve three distinct functions:

(1) It shall be a differential amplifier which amplifies the high impedance receiver test points without reflecting the system power ground to the test points.

(2) It shall amplify the signal from 7 volts input peak-to-peak to 15 volts peak-to-peak output at 50 ohms output impedance.

(3) It shall detect the presence and quality of the AN/SSQ-53 sonobuoy 7.5 kHz pilot tone and indicate the presence of a good signal by supplying a ground path to light an external 28 vdc control lamp.

3.5.12.2 (U) Power Input - 28 vdc from transmitter-multiplexer (±0.5V) at 2 ma.

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*3.5.12.3 (U) Signal Input - 7 volts ac peak-to-peak maximum in 10 Hz to 17.4 kHz band (from DISC and GND test points of the receiver).

*3.5.12.4 (U) Signal Output - 15 volts ac peak-to-peak for ±75 kHz deviation of receiver, at 50 ohms, to multiplexer.

3.5.12.5 (U) Form Factor - The dimensions of the amplifier shall be in accordance with figure 31.

4. QUALITY ASSURANCE PROVISIONS

4.1 (U) Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services and conform to prescribed requirements.

4.1.1 (U) Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

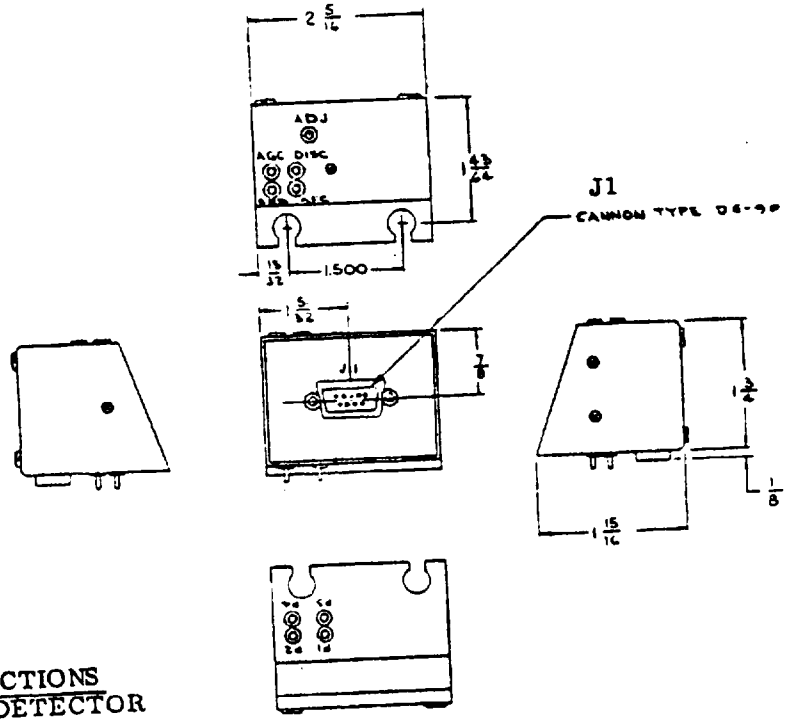
(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 (U) Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of paragraph 4.6. The Government procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 (U) Preproduction (First Article) Test Data - The contractor shall submit test data collected in conducting these tests to the procuring activity for review.

*4.2.2 (U) Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422, interference tests in accordance with MIL-STD-461, and reliability qualification tests in accordance with paragraph 4.4.3.1.

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J1 PIN FUNCTIONS

1	BUOY DETECTOR
2	SPARE
3	SPARE
4	SPARE
5	SIGNAL OUTPUT
6	SPARE
7	+28 VDC
9	+28 VDC RETURN

NOTE:
TOLERANCES UNLESS OTHERWISE SPECIFIED
FRACTIONS: ±1/64
DECIMALS: ±.005

Figure 31. Amplifier AM-6695/AKT-22(V), Form Factor and Connector Requirements

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4.2.3 (U) Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approval preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 (U) Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 (U) Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 (U) Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 (U) Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 (U) Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 (U) Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance as production item.

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4.4 (U) Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be subject to monitoring and witnessing by the government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 (U) Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 (U) Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 (U) Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.1.3 (U) Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours minimum without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3g's

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The equipment shall be vibrated (without vibration isolators) for a period of five minutes prior to the beginning of the ten hour period of operation. The equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration shall be vertical to the normal mounting plane. During the ten hour period of operation following the five minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, and a failure analysis warrants it, the equipment should be repaired and the test started over, except that the vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 (U) Sampling Tests - Equipments selected for sampling test shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1#
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

One out of the first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.4.2.1 (U) Scope of Tests - As a minimum, each equipment selected for sampling test shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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*4.4.3 (U) Reliability Assurance Tests. Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and AR-34. Equipment selected for reliability assurance tests shall have first passed the Individual Test. Failure analysis reports shall be provided for all failures except those occurring during burn-in. Corrective action shall be taken to prevent the reoccurrence of all failures.

*4.4.3.1 (U) Qualification Phase. Prior to the acceptance of equipments under the contract, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Acceptance Reject Criteria for Test Plan III shall be used.

*4.4.3.2 (U) All Equipment Test. Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for a minimum of 74 hours. The last 24 operating hours shall be failure free. If a failure occurs during this 24 hour operating period the equipment shall be repaired, proper operation verified and the test continued. The accumulation of failure free time shall start at zero time upon resumption of testing. Test Level E of MIL-STD-781 shall be used.

a. Prior to the 74 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is used, the details thereof must be included in the approved test procedures. The burn-in shall be the same for each equipment in time and severity.

b. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the Naval Air Systems Command, Attention: Avionics Division.

c. At any time that the current totals of test hours and test failures plotted on Test Plan II curve shows a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other that a reject decision.

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4.4.3.3 (U) Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 (U) Special Tests - Special tests shall be conducted for the purpose of checking the effect of any Class 1 design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 (U) Special Test Schedule - Selection of equipments for special tests shall be made as follows

(1) On an early equipment after engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required, as determined by the procuring activity.

4.4.4.2 (U) Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.5 (U) Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 (U) Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or special tests may be selected for Life Tests. Test Life accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof.

4.5.1 (U) Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	27.5 ±2.0 volts

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4.5.2 (U) Test Periods - The test may run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

*4.5.3 (U) Performance Check - At approximately 24-hour intervals during the test; a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 (U) Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 (U) Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 (U) Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 (U) Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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4.8 (U) Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 (U) Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 (U) General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

*6.1 (U) Intended Use - The AN/AKT-22(V) shall be a multiple airframe variable configuration transmitting set used to relay ASW signals and data to shipboard ASCAC.

6.2 (U) Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of applicable levels of packaging and packing.
(3) Distribution of required operating frequencies.

6.3 (U) Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.
(2) This Specification - This specification shall have precedence over all applicable subsidiary specification. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.
(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 (U) Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 (U) Type Designation - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 (U) Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

*6.7 (U) Associated Equipment -

Radio Receiving Set AN/ARR-52A
Radio Receiving Set AN/ARR-75
Telemetric Data Receiving Set AN/SKR-3A
Telemetric Data Receiving Set AN/SKR-3B
Telemetric Data Receiving Set AN/SKR-4
Telemetric Data Receiving Set AN/SKR-4A
Telemetric Data Receiving Set AN/SKR-5
Sonobuoy AN/SSQ-36
Sonobuoy AN/SSQ-41
Sonobuoy AN/SSQ-47
Sonobuoy AN/SSQ-53

6.8 (U) This specification is under the cognizance of AIR-53341D.

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VIBRATION REQUIREMENTS FOR
TRANSMITTING SET, TELEMETRIC DATA,
AN/AKT-22(V)

The vibration requirements are shown in figure 32.

VIBRATION AMPLITUDE VS. FREQUENCY

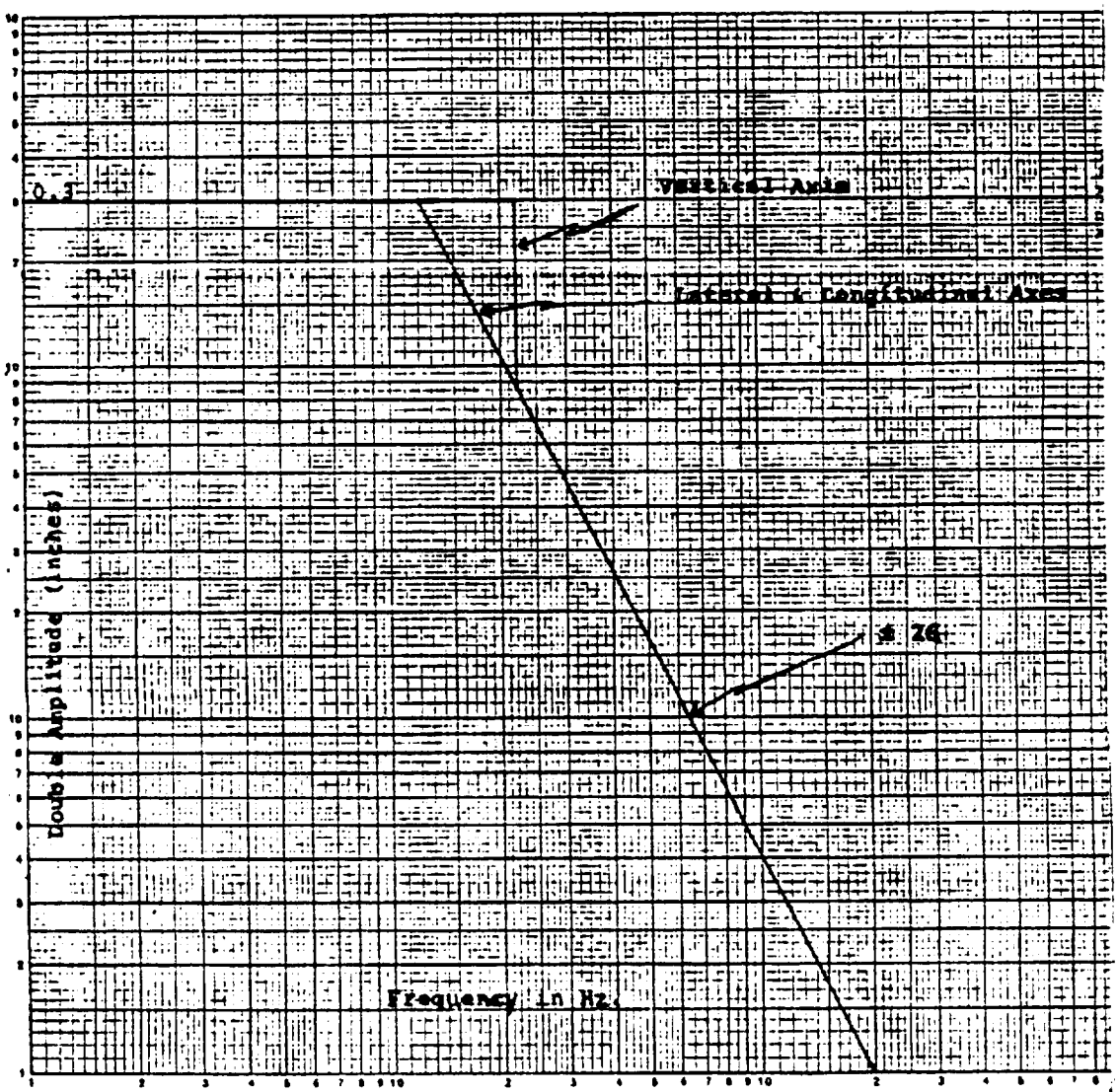


Figure 32 Vibration Requirements

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